

Unified Constraint Framework for Exotic Spin-Dependent Interactions: Matter–Antimatter Sector Comparison

By

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Dedication

To everyone who insists that a symmetry as fundamental as CPT deserves to be checked, not assumed.

Abstract

Exotic spin-dependent interactions mediated by light, weakly coupled bosons (axions, axion-like particles, dark photons) are a well-motivated low-energy signature of physics beyond the Standard Model. Two languages are used to describe them: the Standard Model Extension (SME), a relativistic effective field theory of Lorentz and CPT violation, and the Dobrescu–Mocioiu (DM) classification, a complete basis of sixteen non-relativistic potentials (V_1 – V_{16}) in which experiments actually report their bounds. No complete, systematic dictionary previously connected the two, and no framework existed for using such a dictionary to compare matter-sector and antimatter-sector constraints as a quantitative test of CPT symmetry. This thesis develops both.

The Foldy–Wouthuysen transformation is applied to the SME fermion-sector coefficients b_μ , $d_{\mu\nu}$, $g_{\lambda\mu\nu}$ and $H_{\mu\nu}$ to derive the non-relativistic Hamiltonian each generates and match it onto the corresponding DM potential(s). These four are shown to be the complete set that generates spin-dependent non-relativistic potentials: a_μ , $c_{\mu\nu}$ and e_μ appear only in spin-independent terms and f_μ not at all through third order in $1/m$, so none has an image in the DM basis. All four derivations are obtained in full and agree with established results, including the momentum-dependent d_{ij} and d_{00} components, whose apparent sign discrepancy against the literature is shown to arise from a momentum-index convention in the reference result rather than from the derivation. A dimensionless CPT asymmetry parameter, $A_\alpha = (g_\alpha^f - g_\alpha^{\bar{f}})/(g_\alpha^f + g_\alpha^{\bar{f}})$, is defined and, using a purpose-built pipeline (SPIN-DEP), evaluated against a compiled database of 283 experimental constraint datasets, 247 of which enter the analysis, spanning nitrogen-vacancy-centre magnetometry, torsion balances, atomic co-magnetometers, positronium spectroscopy, and antimatter experiments, standardised to a common interaction-range axis.

Only 22 of the 247 analysed datasets involve an antimatter sector, from which fifteen matter–antimatter pairs could be statistically compared. Every pair shows a significant asymmetry after correcting for autocorrelation among interpolated points, yet an analytic argument, confirmed empirically across all fifteen pairs, shows that A_α built from one-sided experimental upper bounds cannot by itself distinguish genuine CPT violation from an ordinary sensitivity gap between the matter- and antimatter-sector experiments being compared. A gap analysis by fermion sector and by DM potential identifies the e – N , n – N , p – N , e – n , n – p , and n – n sectors, and the V_{1a} , V_{9+10} , and V_{12+13} potentials, as the largest

opportunities for new antimatter-sector measurements. The thesis concludes that the antimatter-sector coverage gap, not statistical power, is the binding constraint on testing CPT symmetry through exotic spin-dependent interactions, and offers the mapping table, the database, and the analysis pipeline as an extensible resource for closing it.

Keywords: Standard Model Extension, Dobrescu–Mocioiu potentials, Foldy–Wouthuysen transformation, CPT symmetry, exotic spin-dependent interactions, matter–antimatter asymmetry.

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Certification

I certify that this work was carried out by Oyewo Temidayo Solomon in the Department of Physics, University of Ibadan, Nigeria.

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Chapter 1

Introduction

1.1 Background to the Research

The Standard Model has been tested to extraordinary precision, yet it is not considered a complete theory because it says nothing about gravity at the quantum level, it has no dark matter candidate despite dark matter making up roughly a quarter of the universe’s energy budget (Planck Collaboration, 2020), and it cannot explain why the universe is full of matter and almost no antimatter. These gaps motivate the search for physics beyond the Standard Model (BSM), one well-motivated form of which is a class of light, weakly coupled bosonic fields, such as axions, axion-like particles, and dark photons, that mediate feeble, long-range *fifth forces*.

This thesis is concerned with a specific subset of fifth forces: those that couple to fermion spin rather than to charge or mass alone. Spin has no classical analogue, which makes *exotic spin-dependent interactions* an unusually clean channel for probing symmetry violation. Two languages are used to describe such interactions. The Standard Model Extension (SME) of Colladay and Kostelecký (1997, 1998) is a relativistic effective field theory that systematically parametrises Lorentz and CPT violation through fixed background coefficients such as b_μ , $H_{\mu\nu}$, and $d_{\mu\nu}$. The Dobrescu–Mocioiu (DM) classification (Dobrescu and Mocioiu, 2006) is a complete, non-relativistic basis of sixteen potentials, V_1 – V_{16} , in which nearly all laboratory searches for exotic forces report their bounds. Connecting the two requires the Foldy–Wouthuysen (FW) transformation (Foldy and Wouthuysen, 1950), the standard technique for extracting the non-relativistic limit of a relativistic Dirac Hamiltonian.

Separately, antimatter physics has become one of the sharpest available tools for testing fundamental symmetries. Experiments at CERN, such as BASE and ALPHA, compare matter and antimatter magnetic moments and spectral lines at a precision that directly tests charge-parity-time (CPT) invariance (Smorra *et al.*, 2017; Ahmadi *et al.*, 2017). Combining these two threads, exotic spin-dependent interaction searches and an-

timatter precision physics, opens a regime in which a CPT-violating interaction would appear as a measurable asymmetry between the coupling constants measured for matter and for antimatter.

1.2 Research Problem and Hypotheses

Despite over a hundred experiments compiled through 2024 (Cong *et al.*, 2025), no unified framework systematically translates between the SME and DM formalisms, and no framework exists for comparing matter-sector and antimatter-sector constraints on exotic spin-dependent interactions within a single, quantitative measure of CPT consistency. The SME describes Lorentz- and CPT-violating physics through operator coefficients that are not directly expressed as measurable potentials; the DM basis is experimentally convenient but does not make its relativistic origin explicit. Prior attempts at a mapping between the two are partial, restricted to subsets of coefficients or potentials, and no compiled constraint database exists in a form that allows a direct, standardised comparison of matter- and antimatter-sector bounds across particle species, interaction types, and experimental platforms.

This research problem yields two working hypotheses examined in this thesis. First, that a complete Foldy–Wouthuysen reduction of the four SME coefficients that generate spin-dependent terms, the CPT-odd b_μ and $g_{\lambda\mu\nu}$ and the CPT-even $H_{\mu\nu}$ and $d_{\mu\nu}$, can be matched, coefficient by coefficient, onto the DM potential basis, and that the resulting potentials will change sign between matter and antimatter for the CPT-odd coefficients but not for the CPT-even ones. Second, that a dimensionless asymmetry parameter A_α , computed from currently published experimental bounds, can be used to test this prediction quantitatively once matter- and antimatter-sector datasets are compiled and standardised.

The first hypothesis is confirmed for all four coefficients, with one clarification about what is being compared. The sign flip for the CPT-odd b_μ and $g_{\lambda\mu\nu}$, and its absence for the CPT-even $H_{\mu\nu}$ and $d_{\mu\nu}$, hold between CPT-conjugate states, in which the antiparticle’s spin is reversed. At the same physical spin the pattern is reversed, as Kostelecký and Lane (1999b) state (Section 4.1.1). The second hypothesis is limited by the nature of the data. Published bounds are one-sided upper limits rather than signed measurements, so A_α alone cannot always separate a genuine CPT-violating signal from an ordinary gap in experimental sensitivity (Chapters 3 and 4). This thesis treats that limitation as a central finding rather than an incidental caveat.

1.3 Justification for the Research

A unified SME–DM mapping gives experimentalists a direct way to translate a measured or constrained potential into a bound on a relativistic, symmetry-classified coefficient, and vice versa, rather than relying on partial, coefficient-specific translations scattered across the literature. A standardised, cross-platform constraint database removes the unit and convention inconsistencies that currently make it difficult to compare bounds from nitrogen-vacancy-centre magnetometry, torsion balances, atomic co-magnetometers, positronium spectroscopy, and antimatter spectroscopy on equal footing. Together, these tools enable the first systematic matter–antimatter comparison of spin-dependent forces and provide a reusable, extensible pipeline (SPINDEP) rather than a one-off calculation, one that automatically incorporates future antimatter-sector measurements as they are published. The results are relevant to ongoing experimental programmes at CERN (BASE, ALPHA), to precision quantum sensing laboratories, and to the SME data tables maintained by Kostelecký and Russell (2011), which this work is intended to complement.

1.4 Aims and Objectives

The primary objective of this research is to develop a unified theoretical and computational framework that connects Standard Model Extension coefficients to Dobrescu–Mocioiu spin-dependent potentials and enables systematic comparison of matter- and antimatter-sector constraints to test CPT symmetry. The specific objectives are:

1. to derive explicit mappings between the SME fermion-sector coefficients b_μ , $H_{\mu\nu}$, $d_{\mu\nu}$, and $g_{\lambda\mu\nu}$ and the Dobrescu–Mocioiu coupling structures g_s , g_p , g_V , and g_A using the Foldy–Wouthuysen transformation;
2. to construct a unified, standardised database of experimental constraints from nitrogen-vacancy-centre magnetometry, torsion balances, atomic co-magnetometers, positronium spectroscopy, and antimatter experiments;
3. to define and compute the CPT asymmetry parameter A_α across every matter–antimatter pair the compiled database can support;
4. to perform a global coverage analysis identifying unconstrained or weakly constrained regions of the parameter space, particularly in the antimatter sector; and
5. to propose concrete experimental and data-compilation strategies for closing the identified gaps.

1.5 Outline of the Thesis

- Chapter 1 introduces the motivation, research questions, and objectives of the thesis.
- Chapter 2 reviews the theoretical background, the experimental literature, and the specific inadequacies in existing knowledge that this thesis addresses.
- Chapter 3 describes the materials and methods: the Foldy–Wouthuysen procedure used to derive the SME–DM mapping, and the database compilation and statistical methodology used to compute the CPT asymmetry parameter.
- Chapter 4 presents the results, the derived coefficient-to-potential mapping, the compiled 283-dataset constraint database, the matter–antimatter asymmetry analysis, and the coverage gap analysis, together with their discussion.
- Chapter 5 presents the conclusions of the thesis and outlines its principal limitations and directions for future work.

Chapter 2

Literature Review

2.1 Theoretical Background

2.1.1 The Standard Model Extension

The Standard Model Extension (SME) is constructed as the most general effective field theory Lagrangian obtainable by adding to the Standard Model (and General Relativity) every observer-scalar term formed by contracting Standard Model field operators with coefficients carrying free Lorentz indices (Colladay and Kostelecký, 1997, 1998). A distinction central to the framework is between two kinds of Lorentz transformation. *Observer* transformations rotate or boost the coordinate frame used to describe a system without touching the system itself; the SME is constructed to respect observer Lorentz invariance exactly. *Particle* transformations rotate or boost a localised particle or field configuration while holding the background coefficients fixed, and it is only under this class of transformation that the SME's background coefficients act as fixed, non-dynamical tensors that break the symmetry. This distinction is what allows the SME to remain a coordinate-independent, internally consistent field theory while still describing genuine, in-principle observable violations of Lorentz invariance.

The SME Lagrangian is written schematically as

$$\mathcal{L}_{\text{SME}} = \mathcal{L}_{\text{SM}} + \delta\mathcal{L}_{\text{LV}}, \quad (2.1)$$

where $\mathcal{L}_{\text{SM}}$ is the ordinary Standard Model Lagrangian and $\delta\mathcal{L}_{\text{LV}}$ collects every observer-Lorentz-scalar, particle-Lorentz-violating operator that can be built from Standard Model fields. This thesis works within the *minimal* SME, the subset of $\delta\mathcal{L}_{\text{LV}}$ of renormalisable mass dimension (≤ 4) in the fermion sector, where existing spin-dependent exotic-interaction experiments place their bounds.

For a single Dirac fermion ψ of mass m , the general minimal SME fermion-sector

Lagrangian is

$$\mathcal{L}_{\text{fermion}} = \frac{i}{2} \bar{\psi} \Gamma^\nu \overleftrightarrow{D}_\nu \psi - \bar{\psi} M \psi, \quad (2.2)$$

where D_ν is the gauge-covariant derivative, $\overleftrightarrow{D}_\nu \equiv D_\nu - \overleftarrow{D}_\nu$, and the Lorentz-violating physics is packaged entirely into two composite matrix-valued objects,

$$\Gamma^\nu = \gamma^\nu + c^{\mu\nu} \gamma_\mu + d^{\mu\nu} \gamma_5 \gamma_\mu + e^\nu + i f^\nu \gamma_5 + \frac{1}{2} g^{\lambda\mu\nu} \sigma_{\lambda\mu}, \quad (2.3)$$

$$M = m + a_\mu \gamma^\mu + b_\mu \gamma_5 \gamma^\mu + \frac{1}{2} H_{\mu\nu} \sigma^{\mu\nu}. \quad (2.4)$$

Γ^ν modifies the kinetic (derivative-coupling) sector, while M modifies the mass sector. Every coefficient in Eqs. (2.3) and (2.4) is a fixed background tensor, not a dynamical field, and each is in principle independent for every Standard Model fermion species. Setting all coefficients to zero except m recovers the ordinary Dirac Lagrangian. For the leading-order (dimension ≤ 4) subset used throughout this thesis, Eq. (2.2) reduces to the compact form

$$\begin{aligned} \mathcal{L}_{\text{fermion}} = & \bar{\psi} (i \gamma^\mu D_\mu - m) \psi - a_\mu \bar{\psi} \gamma^\mu \psi - b_\mu \bar{\psi} \gamma_5 \gamma^\mu \psi \\ & - \frac{1}{2} H_{\mu\nu} \bar{\psi} \sigma^{\mu\nu} \psi - c_{\mu\nu} \bar{\psi} \gamma^\mu D^\nu \psi - d_{\mu\nu} \bar{\psi} \gamma_5 \gamma^\mu D^\nu \psi, \end{aligned} \quad (2.5)$$

where D_μ is the gauge-covariant derivative and $\sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu]$. Note the operator ordering in the b_μ term, $\gamma_5 \gamma^\mu$ rather than $\gamma^\mu \gamma_5$: since $\{\gamma_5, \gamma^\mu\} = 0$, the two orderings differ by an overall sign, so this convention fixes the sign of every non-relativistic result derived from b_μ later in this thesis. Coefficients in the mass sector M (a_μ , b_μ , $H_{\mu\nu}$) have dimensions of mass, while those in the kinetic sector Γ^ν are dimensionless (Kostelecký and Lane, 1999a).

Equation (2.5) omits e_μ , f_μ and $g_{\lambda\mu\nu}$. These three are also dimensionless and renormalisable in the single-fermion Lagrangian, but they cannot come from any renormalisable term of the full standard-model extension, because every such term would violate its $\text{SU}(2) \times \text{U}(1)$ gauge structure. They could arise only from non-renormalisable, higher-dimensional operators involving the Higgs field, and are therefore expected to be strongly suppressed (Colladay and Kostelecký, 1998). For the electron, they are absent or suppressed relative to the other coefficients. For the proton and neutron, however, strong binding inside a composite particle could generate appreciable effective values (Kostelecký and Lane, 1999a). Of the three, only $g_{\lambda\mu\nu}$ produces spin-dependent terms. e_μ enters the non-relativistic Hamiltonian only through spin-independent terms, and f_μ does not enter it at all (Kostelecký and Lane, 1999a, 1999b). f_μ can in fact be absorbed into the other coefficients by a field redefinition (Altschul, 2006). This thesis therefore keeps $g_{\lambda\mu\nu}$ and sets e_μ and f_μ aside. Table 2.1 classifies all eight coefficients by CPT parity, mass dimension and leading spin-dependent role.

The CPT parity of a coefficient follows from how its associated fermion bilinear $\bar{\psi} \Gamma \psi$

Coefficient	CPT	Mass dim.	Leading spin-dependent role
a_μ	Odd	1	None at leading order; velocity-dependent at $O(1/m)$
b_μ	Odd	1	Zeeman-like spin coupling, V_2
$c_{\mu\nu}$	Even	0	None; direction-dependent kinetic modification, spin-independent
$d_{\mu\nu}$	Even	0	Spin-velocity and mass-enhanced couplings, V_2, V_7, V_8
e_μ	Odd	0	None; energy shift and velocity-dependent term, spin-independent
f_μ	Odd	0	None; absent from the non-relativistic Hamiltonian at linear order and removable by a field redefinition
$g_{\lambda\mu\nu}$	Odd	0	Spin couplings via $\sigma^{\lambda\mu}$, V_2, V_7, V_8
$H_{\mu\nu}$	Even	1	Anisotropic spin coupling, V_3, V_7

Table 2.1: Classification of minimal (and near-minimal) SME fermion-sector coefficients (Colladay and Kostelecký, 1998; Kostelecký and Lane, 1999a, 1999b). “Mass dim.” is the mass dimension of the coefficient itself; the operators it multiplies have dimension 3 in the mass sector and 4 in the kinetic sector.

transforms under the combined charge conjugation, parity, and time-reversal operation,

$$\text{CPT}[\bar{\psi} \Gamma \psi] = \eta_\Gamma \bar{\psi} \Gamma \psi, \quad \eta_\Gamma = \begin{cases} +1 & \Gamma \in \{\not{1}, \gamma^\mu, \sigma^{\mu\nu}\} \\ -1 & \Gamma \in \{\gamma_5, \gamma_5 \gamma^\mu\} \end{cases}, \quad (2.6)$$

and the corresponding non-relativistic Hamiltonians obey the general rule

$$H_{NR}^{\bar{f}}(X^{\text{CPT-odd}}) = -H_{NR}^f(X^{\text{CPT-odd}}), \quad H_{NR}^{\bar{f}}(X^{\text{CPT-even}}) = +H_{NR}^f(X^{\text{CPT-even}}). \quad (2.7)$$

2.1.2 The Dobrescu–Mocioiu Classification of Exotic Potentials

Experimental searches for new long-range forces are conducted in the non-relativistic laboratory frame, not in the relativistic language of the SME. Dobrescu and Mocioiu (2006) closed this gap by enumerating, from first principles, every non-relativistic potential that can arise from the exchange of a single boson of arbitrary spin and parity between two spin- $\frac{1}{2}$ fermions. The result is a complete basis of sixteen potentials,

$$V(\mathbf{r}) = \sum_{i=1}^{16} V_i(\mathbf{r}), \quad (2.8)$$

parametrised by coupling constants g_s, g_p, g_V, g_A (scalar, pseudoscalar, vector, axial-vector). With $y(r) = e^{-r/\lambda}/(4\pi)$, $\lambda = 1/m_\phi$ the mediator's Compton wavelength, $\hat{\mathbf{r}}$ pointing from particle 2 to particle 1, and $\mathbf{v}$ the relative velocity of the pair, the sixteen potentials are (Cong *et al.*, 2025; Dobrescu and Mocioiu, 2006):

$$V_1 = \frac{1}{r} y(r) \quad (2.9)$$

$$V_2 = \frac{1}{r} (\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2) y(r) \quad (2.10)$$

$$V_3 = \frac{e^{-r/\lambda}}{4\pi m^2} \left\{ \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \left[\frac{1}{r^3} + \frac{1}{\lambda r^2} \right] - (\boldsymbol{\sigma}_1 \cdot \hat{\mathbf{r}})(\boldsymbol{\sigma}_2 \cdot \hat{\mathbf{r}}) \left[\frac{3}{r^3} + \frac{3}{\lambda r^2} + \frac{1}{\lambda^2 r} \right] \right\} \quad (2.11)$$

$$V_{4,5} = -\frac{1}{2mr^2} (\boldsymbol{\sigma}_1 \pm \boldsymbol{\sigma}_2) \cdot (\mathbf{v} \times \hat{\mathbf{r}}) \left(1 - r \frac{d}{dr} \right) y(r) \quad (2.12)$$

$$V_{6,7} = -\frac{1}{2mr^2} \left[(\boldsymbol{\sigma}_1 \cdot \mathbf{v})(\boldsymbol{\sigma}_2 \cdot \hat{\mathbf{r}}) \pm (\boldsymbol{\sigma}_1 \cdot \hat{\mathbf{r}})(\boldsymbol{\sigma}_2 \cdot \mathbf{v}) \right] \left(1 - r \frac{d}{dr} \right) y(r) \quad (2.13)$$

$$V_8 = \frac{1}{r} (\boldsymbol{\sigma}_1 \cdot \mathbf{v})(\boldsymbol{\sigma}_2 \cdot \mathbf{v}) y(r) \quad (2.14)$$

$$V_{9,10} = -\frac{1}{2mr^2} (\boldsymbol{\sigma}_1 \pm \boldsymbol{\sigma}_2) \cdot \hat{\mathbf{r}} \left(1 - r \frac{d}{dr} \right) y(r) \quad (2.15)$$

$$V_{11} = -\frac{1}{mr^2} (\boldsymbol{\sigma}_1 \times \boldsymbol{\sigma}_2) \cdot \hat{\mathbf{r}} \left(1 - r \frac{d}{dr} \right) y(r) \quad (2.16)$$

$$V_{12,13} = \frac{1}{2r} (\boldsymbol{\sigma}_1 \pm \boldsymbol{\sigma}_2) \cdot \mathbf{v} y(r) \quad (2.17)$$

$$V_{14} = \frac{1}{r} (\boldsymbol{\sigma}_1 \times \boldsymbol{\sigma}_2) \cdot \mathbf{v} y(r) \quad (2.18)$$

$$V_{15} = -\frac{3}{2m^2 r^3} \left\{ [\boldsymbol{\sigma}_1 \cdot (\mathbf{v} \times \hat{\mathbf{r}})](\boldsymbol{\sigma}_2 \cdot \hat{\mathbf{r}}) + (\boldsymbol{\sigma}_1 \cdot \hat{\mathbf{r}})[\boldsymbol{\sigma}_2 \cdot (\mathbf{v} \times \hat{\mathbf{r}})] \right\} \left(1 - r \frac{d}{dr} + \frac{r^2}{3} \frac{d^2}{dr^2} \right) y(r) \quad (2.19)$$

$$V_{16} = -\frac{1}{2mr^2} \left\{ [\boldsymbol{\sigma}_1 \cdot (\mathbf{v} \times \hat{\mathbf{r}})](\boldsymbol{\sigma}_2 \cdot \mathbf{v}) + (\boldsymbol{\sigma}_1 \cdot \mathbf{v})[\boldsymbol{\sigma}_2 \cdot (\mathbf{v} \times \hat{\mathbf{r}})] \right\} \left(1 - r \frac{d}{dr} \right) y(r) \quad (2.20)$$

Table 2.2 groups the sixteen potentials by structural class, the organising principle Dobrescu and Mocioiu use to enumerate them exhaustively, and the property that determines which SME coefficient(s) can generate a given V_i (Chapter 4).

Structural class	Potentials
Monopole–monopole (spin-independent)	V_1
Monopole–dipole	V_9, V_{10}
Dipole–dipole (spin–spin)	V_2, V_3
Spin–velocity	V_4, V_5, V_6, V_7
Velocity–velocity	V_8
Spin cross-product	V_{11}
Single-spin–velocity	V_{12}, V_{13}, V_{14}
Higher-order spin–velocity	V_{15}, V_{16}

Table 2.2: Structural classification of the Dobrescu–Mocioiu potentials.

2.1.3 The Foldy–Wouthuysen Transformation

The Foldy–Wouthuysen transformation is selected over alternative non-relativistic reduction methods for two reasons specific to this thesis. The Klein–Gordon equation is excluded outright: it describes spin-0 fields and carries no spinor structure, making it incapable of generating the spin-operator combinations $\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$, $\boldsymbol{\sigma} \cdot \hat{r}$, and $\boldsymbol{\sigma} \cdot \mathbf{v}$ that define the V_1 – V_{16} basis. Working directly with the full Dirac equation, while formally correct, leaves the result in a relativistic form incompatible with the non-relativistic experimental observables the DM potentials are written for; the FW transformation resolves this by producing a block-diagonal Hamiltonian in which the upper block is immediately readable as a non-relativistic Pauli-operator expression, order by order in $1/m$, ready for direct matching against V_i .

The Foldy–Wouthuysen (FW) transformation (Foldy and Wouthuysen, 1950) is the standard technique for systematically extracting the non-relativistic limit of a relativistic Dirac Hamiltonian, order by order in $1/m$, while keeping every step exact and unitary. It decouples the upper (particle) and lower (antiparticle) blocks of the Dirac spinor by eliminating, order by order, the operators that mix the two blocks. Writing $H = \beta m + \varepsilon_{\text{even}} + \varepsilon_{\text{odd}}$, where $\varepsilon_{\text{even}}$ commutes with $\beta = \gamma^0$ and ε_{odd} anticommutes with it, the transformed Hamiltonian expands to order $1/m$ as

$$H' \approx \beta m + \varepsilon_{\text{even}} + \frac{\beta \varepsilon_{\text{odd}}^2}{2m} - \frac{[\varepsilon_{\text{odd}}, \varepsilon_{\text{even}}]}{4m} + O(m^{-2}). \quad (2.21)$$

An even piece of the original perturbation survives into the non-relativistic Hamiltonian unchanged at order m^0 ; an odd piece contributes only at order $1/m$, through the $\beta \varepsilon_{\text{odd}}^2/2m$ term. This is why coefficients coupling through odd operators (such as b_0 and H_{0i}) are systematically subleading relative to those coupling through even operators (such as b_i and H_{ij}). Projecting the block-diagonal result onto the upper block gives the matter-sector Hamiltonian directly. The antiparticle sector follows from the lower (negative-energy) block by the hole-theory reinterpretation, which reverses the sign of the energy, the momentum and the spin. Charge conjugation of the bilinear alone cannot supply the matter–antimatter sign difference for b_μ , since $\bar{\psi} \gamma_5 \gamma^\mu \psi$ is even under C for every μ . The difference is a CPT effect, and it appears when CPT-conjugate states are compared (Chapter 4).

2.1.4 CPT Symmetry and the Asymmetry Parameter

CPT symmetry, invariance under the combined transformation of charge conjugation, parity, and time reversal, is a theorem of local, Lorentz-invariant quantum field theory (Colladay and Kostelecký, 1997). To quantify possible CPT asymmetries in exotic inter-

actions, this thesis defines the dimensionless asymmetry parameter

$$A_\alpha = \frac{g_\alpha^f - g_\alpha^{\bar{f}}}{g_\alpha^f + g_\alpha^{\bar{f}}}, \quad \alpha \in \{s, p, V, A\}, \quad (2.22)$$

where g_α^f and $g_\alpha^{\bar{f}}$ are the effective coupling constants for a fermion species and its antifermion in interaction channel α . The antifermion coupling refers to the CPT-conjugate state, with the same momentum and reversed spin. With this definition, exact CPT symmetry gives $A_\alpha = 0$ for every channel. As Chapter 4 derives in detail, g_α^f and $g_\alpha^{\bar{f}}$ as actually reported in the literature are one-sided experimental upper bounds rather than signed measured couplings, which limits how directly a non-zero A_α can be read as evidence of CPT violation.

2.2 Review of Existing Knowledge

2.2.1 Experimental Platforms for Exotic Spin-Dependent Interactions

The search for exotic spin-dependent interactions spans a diverse range of experimental platforms, each optimised for a different length scale, particle species, and sensitivity regime. Nitrogen-vacancy (NV) centres in diamond are sensitive solid-state quantum sensors for weak spin-dependent fields at sub-micrometre scales, owing to long coherence times and optical spin readout (Rong *et al.*, 2018a, 2018b), and are effective for short-range Yukawa-type potentials from $0.1 \mu\text{m}$ to $50 \mu\text{m}$. Torsion-pendulum experiments provide macroscopic sensitivity to weak spin-dependent and spin-independent forces over mesoscopic distances, typically $10 \mu\text{m}$ to 10cm , using polarised test masses against unpolarised attractors (Heckel *et al.*, 2008). Atomic magnetometers operating in the spin-exchange relaxation-free regime achieve sub-femtotesla sensitivity to exotic spin couplings and are especially relevant for constraining b_μ and $d_{\mu\nu}$ (Vasilakis *et al.*, 2009). Antimatter experiments at CERN provide a direct test of CPT symmetry: BASE measures the antiproton magnetic moment to parts-per-billion precision (Smorra *et al.*, 2017), and ALPHA performs precision spectroscopy on trapped antihydrogen (Ahmadi *et al.*, 2017). Positronium, a purely leptonic bound system free of hadronic uncertainties, provides sensitivity to electron-positron spin couplings at atomic scales, with recent improvements from hyperfine and fine-structure spectroscopy (Fadeev *et al.*, 2022; Adkins, Cassidy and Pérez-Ríos, 2022).

2.2.2 Theoretical Development History

Moody and Wilczek (1984) gave the first systematic treatment of long-range spin-dependent forces mediated by axion-like particles, establishing the Yukawa-type potential structure and the $g_s g_p$ coupling dependence that remains standard today. Colladay and Kostelecký (1997, 1998) constructed the SME as the most general effective field theory allowing Lorentz and CPT violation, and Kostelecký and Lane (1999a, 1999b) derived the non-relativistic limit of SME fermion-sector terms, providing the first formal bridge toward experimentally accessible observables. Dobrescu and Mocioiu (2006) then provided the complete classification of non-relativistic potentials described in Section 2.1.2. More recently, Fadeev *et al.* (2019, 2022) unified exotic spin-dependent interactions under a common effective field theory language and extended the analysis to observables in simple atomic systems, while Cong *et al.* (2025) compiled the most complete modern survey of experimental constraints and exotic interaction classifications through 2024, without, however, constructing a systematic SME–Dobrescu–Mocioiu mapping or a matter–antimatter comparison framework.

2.3 Problems and Inadequacies in Existing Knowledge

Despite these theoretical and experimental advances, the field remains fragmented across multiple formalisms and experimental reporting conventions, which limits cross-platform comparability and prevents a coherent programme of CPT tests using exotic interactions. The SME provides a fully relativistic description through well-defined operator coefficients, but these coefficients are not directly expressed in terms of experimentally measurable potentials. The Dobrescu–Mocioiu framework provides an experimentally convenient basis, but is phenomenological and does not make its connection to relativistic coefficients explicit. Prior attempts at bridging the two via Foldy–Wouthuysen reduction are incomplete, typically restricted to a subset of SME coefficients or a limited class of potentials; a fully general mapping between the complete SME fermion sector and the full V_1 – V_{16} basis has not previously been established. Existing experimental compilations, including the recent and extensive one of Cong *et al.* (2025), are not expressed within a unified parameter space that allows direct comparison across particle species, interaction types, and experimental platforms. Most critically, no existing framework defines a universal CPT asymmetry parameter across all exotic spin-dependent interaction channels, nor has such a parameter been systematically evaluated against compiled experimental data. This prevents a quantitative, model-independent assessment of CPT violation in spin-dependent forces, and is the gap this thesis addresses.

2.4 Limitations of Existing Approaches

Three specific limitations of the existing literature bear directly on the work of this thesis. First, published compilations report constraints as one-sided upper bounds, not signed central values with symmetric uncertainties; any CPT-comparison framework built on such data inherits this limitation and must, as shown in Chapter 4, treat it explicitly rather than assume it away. Second, antimatter-sector measurements are far rarer and far less standardised in units and reporting convention than matter-sector measurements, which existing compilations do not systematically quantify sector by sector or potential by potential. Third, the SME–Dobrescu–Mocioiu correspondence has, prior to this work, only been worked out piecemeal, coefficient by coefficient, in different papers using different conventions for signs, index placement, and normalisation, which makes an independent, internally consistent re-derivation, of the kind undertaken in Chapters 3 and 4 of this thesis, necessary before a reliable matching table can be trusted for quantitative use.

Chapter 3

Materials and Methods

This is a theoretical and computational thesis: its "materials" are the published SME and Dobrescu–Mocioiu (DM) formalisms reviewed in Chapter 2 and the compiled experimental constraint literature, and its principal research tool is SPINDEP, a purpose-built analysis pipeline (Python, NumPy/SciPy, and symbolic computation in SymPy) developed for this thesis. This chapter describes the research procedures used to derive the SME→DM coefficient mapping (Section 3.1) and the methods of analysis used to compile and statistically compare experimental constraints (Section 3.2). The results these procedures produce are presented and discussed in Chapter 4.

3.1 Research Procedures: Foldy–Wouthuysen Reduction of SME Coefficients

Each SME fermion-sector coefficient of interest, b_μ , $H_{\mu\nu}$, $d_{\mu\nu}$, and $g_{\lambda\mu\nu}$, was reduced to its non-relativistic Hamiltonian and matched onto the Dobrescu–Mocioiu potential catalogue by the same five-step procedure.

1. **Write the covariant Lagrangian term.** The relevant term is isolated from the general minimal SME Lagrangian (Eq. 2.5), and its CPT and Lorentz transformation properties are checked directly from the transformation of the corresponding fermion bilinear (or, for the derivative-coupled coefficient $d_{\mu\nu}$, stated following Kostelecký and Lane, 1999a, since the bilinear rule alone does not apply to a derivative-coupled operator).
2. **Derive the correction to the Dirac Hamiltonian.** Varying the modified Lagrangian and solving the resulting modified Dirac equation for $H\psi = E\psi$ gives the coefficient's contribution to the free Dirac Hamiltonian $H_0 = \boldsymbol{\alpha} \cdot \mathbf{p} + \beta m$, in the Dirac representation of the gamma matrices ($\gamma^0 = \text{diag}(1, 1, -1, -1)$, γ_5 off-diagonal).

3. **Classify each contribution as even or odd.** Using standard Dirac-algebra identities ($\{\gamma_5, \gamma^\mu\} = 0$ and the explicit 4×4 matrix forms of $\gamma^0 \gamma_5 \gamma^i$, $\gamma^0 \sigma^{ij}$, and related combinations, verified symbolically with a purpose-written Dirac-algebra module), each component of the correction is classified as block-diagonal (*even*, commuting with β) or block-off-diagonal (*odd*, anticommuting with β).
4. **Apply the Foldy–Wouthuysen expansion.** An even component contributes to the non-relativistic Hamiltonian unchanged at order m^0 , requiring no further transformation. An odd component is combined with the leading odd operator $\boldsymbol{\alpha} \cdot \mathbf{p}$ through the general order- $1/m$ FW expansion (Eq. 2.21), producing a sub-leading, momentum-dependent contribution. This procedure is preferred over a direct equation-of-motion expansion in the full four-component spinor because it delivers the particle-sector Hamiltonian as an explicit 2×2 Pauli expression whose operator structure can be matched term-by-term against the closed-form V_i potentials of Eqs. 2.9–2.20. The antiparticle Hamiltonian is then obtained from the negative-energy (lower) block by the hole-theory reinterpretation: an antiparticle of momentum $\mathbf{p}$ and spin $\mathbf{s}$ is the absence of a negative-energy solution of momentum $-\mathbf{p}$ and spin $-\mathbf{s}$, with the opposite energy. The particle is compared with its CPT-conjugate antiparticle, of the same momentum and reversed spin, which gives $H_{NR}^f(\mathbf{p}, \boldsymbol{\sigma}) = -H_{\text{lower}}(-\mathbf{p}, \boldsymbol{\sigma})$. This route was adopted in preference to a charge-conjugation argument on the bilinear $\bar{\psi} \gamma_5 \gamma^\mu \psi$ after that bilinear was verified, with the same Dirac-algebra tooling, to be even under C for every μ and therefore unable by itself to produce a sign difference. The comparison was verified for every coefficient family against each coefficient’s CPT parity and against Kostelecký and Lane’s (1999b) antiparticle rule at fixed spin.
5. **Match to the Dobrescu–Mocioiu catalogue.** The resulting single-particle non-relativistic Hamiltonian is promoted to a two-body potential by pairing the same vertex structure on both interacting fermions, and the resulting operator structure (in $\boldsymbol{\sigma}_1$, $\boldsymbol{\sigma}_2$, $\hat{\mathbf{r}}$, and $\mathbf{v}$) is compared directly against the closed forms of V_1 – V_{16} (Eqs. 2.9–2.20) to identify which potential(s) the coefficient generates and at what order in $1/m$.

Every derivation obtained by this procedure was additionally verified by an independent, executable symbolic computation (SymPy notebooks `FW_bmu_term.ipynb`, `FW_Hmunu_term.ipynb`, and `FW_dmunu_term.ipynb`), including, for the CPT asymmetry parameter specifically, a symbolic limit calculation used to check the behaviour of Eq. (2.22) under an exact signed CPT-odd substitution (Chapter 4, Section 4.1.3). Where a derived result could be checked against an independently published non-relativistic Hamiltonian (Kostelecký and Lane, 1999a, Eq. 4), this comparison was carried out

component by component rather than only at the level of overall structure. That is how an apparent sign discrepancy in the d_{ij} and d_{00} components (Chapter 4) was first identified, and subsequently traced to the momentum-index convention of the reference result rather than to the derivation itself.

3.2 Methods of Analysis

3.2.1 Constraint Database Compilation

Each experimental constraint used in this thesis is stored as a two-column comma-separated value (CSV) file (interaction range λ in metres, upper bound on the coupling $|g|$), organised in a directory tree by coupling family and interaction class, following the naming convention `{V}{Author}{Year}_m_abs_{sector}.csv`. A parser recursively discovers every CSV file under the dataset root and extracts the potential label, source citation, and fermion sector from the filename and directory path, resolving alternate spellings of antimatter sectors (for example **ebare**, **ebar**, and **epbare** all resolve to the canonical labels **eebar** and **epbar**). Files whose sector or potential token cannot be resolved are retained in the compiled registry with the corresponding field recorded as **UNKNOWN** rather than being silently dropped, so that any classification gap remains visible rather than hidden by omission. Such files are then held out of the physics analysis by an explicit, per-file exclusion list that records a stated reason for each, rather than by a blanket rule on the **UNKNOWN** label; the registry therefore reports both what was compiled and what was analysed, and the difference is auditable. Interaction ranges reported in the literature in a mix of units (metres, MeV^{-1} , eV^{-1} , centimetres, nanometres, femtometres) are converted to a common metre scale by a unit-detection routine that inspects each filename for an explicit unit token before the dataset is used in any downstream analysis. Two datasets are matched into a matter–antimatter pair only when they share the same coupling family, the same DM potential label, and the same interaction class, and their fermion sectors are physically conjugate (for example **ee** with **eebar**), with exactly one of the two flagged as an antimatter dataset; this pairing logic is deliberately conservative and will not match two datasets differing in potential classification even where their sectors are conjugate.

3.2.2 Statistical Methodology for the Asymmetry Analysis

For each matched matter–antimatter pair, both constraint curves are interpolated (log-linear in $\log \lambda$ – $\log g$) onto a common log-spaced grid of 300 points spanning their overlapping λ range, and the asymmetry parameter $A_\alpha(\lambda)$ of Eq. (2.22) is evaluated at every grid point. Statistical significance is assessed in four stages, each addressing a limitation of the one before it.

1. A *uniform χ^2 test* assumes a flat 10% fractional uncertainty at every grid point.
2. A *weighted χ^2 test* instead estimates a per-point fractional uncertainty from the local curvature of each constraint curve in log-log space, on the grounds that a smoothly varying published bound is better determined than one with sharp local features.
3. Both tests above treat the 300 grid points as independent degrees of freedom, which is not justified, since the points are interpolated from a much smaller number of real measurements along smooth curves and adjacent points are strongly correlated. An *effective degrees of freedom* is therefore estimated from the autocorrelation length τ of the χ^2 residuals in grid-index space (the lag at which the residual autocorrelation first drops below $1/e$), as $n_{\text{eff}} = n/\tau$. The weighted χ^2 is scaled down by the same factor, $\chi_{\text{eff}}^2 = \chi^2 n_{\text{eff}}/n$, and compared with a χ^2 distribution on n_{eff} degrees of freedom. Since most of the resulting p -values lie far below double precision, the significance is quoted as the equivalent one-sided Gaussian Z_{eff} .
4. A *parametric bootstrap* (2000 resamples, propagating the estimated per-point uncertainty through to A_α itself) gives a 95% confidence interval on the mean $|A_\alpha|$ for each pair.

Two points about this bootstrap step matter for how its intervals are read. First, each resample perturbs both interpolated couplings by their estimated fractional uncertainties (log-normally, so they stay positive) and recomputes A_α exactly. Because $|A_\alpha|$ is nonlinear in the couplings, folded at zero and saturating at one, the resampled means sit slightly off the point estimate even when the perturbations are symmetric, and that offset, estimated as the median bootstrap bias, is removed before the interval is quoted. Second, the interval only reflects the spread implied by the assumed curve uncertainties; for $|A_\alpha|$ an interval that excludes zero is not, on its own, evidence of a non-zero asymmetry.

This methodology is applied uniformly to every matched pair in the compiled database (Chapter 4), and its results are interpreted throughout in light of the analytic sensitivity-gap argument developed in Section 2.1.4 and Chapter 4: statistical significance of an asymmetry is a separate claim from evidence of CPT violation, and the methodology above establishes the former without, on its own, establishing the latter.

Chapter 4

Results and Discussion

This chapter presents the results obtained by the procedures of Chapter 3: the Foldy–Wouthuysen (FW) mapping from SME coefficients to Dobrescu–Mocioiu (DM) potentials (Section 4.1), the compiled experimental constraint database (Section 4.2), the matter–antimatter asymmetry analysis (Section 4.3), and the resulting coverage gap analysis (Section 4.4). Each result is discussed as it is presented, since the physical interpretation of every numerical result in this thesis depends on the sensitivity-gap caveat established in Section 4.1.1 below.

4.1 Foldy–Wouthuysen Derivation Results

4.1.1 The b_μ Term

The CPT-odd term $\mathcal{L}_b = -b_\mu \bar{\psi} \gamma_5 \gamma^\mu \psi$ splits into a spatial part b_i , which is even and therefore exact at order m^0 , and a temporal part b_0 , which is odd and enters only at order $1/m$. Direct computation in the Dirac representation gives $\gamma^0 \gamma_5 \gamma^i = -\Sigma^i$, so that, projected onto the upper block,

$$H_{NR}(b_i) = -\mathbf{b} \cdot \boldsymbol{\sigma} \quad (\text{matter}), \quad (4.1)$$

already in non-relativistic form with no further FW iteration required. The temporal component is odd, since $\gamma^0 \gamma_5 \gamma^0 = -\gamma_5$ (using $\{\gamma_5, \gamma^0\} = 0$ and $(\gamma^0)^2 = I$), so it does not contribute at m^0 and instead enters through the FW cross-term between the free odd operator $\boldsymbol{\alpha} \cdot \mathbf{p}$ and the perturbing odd operator $\mathcal{O}_{b_0} = -b_0 \gamma_5$: expanding $(\boldsymbol{\alpha} \cdot \mathbf{p} + \mathcal{O}_{b_0})^2$ and keeping only the piece linear in b_0 ,

$$H_{NR}(b_0) = \frac{\beta}{2m} \{ \boldsymbol{\alpha} \cdot \mathbf{p}, \mathcal{O}_{b_0} \} \Big|_{\text{upper block}} = -\frac{b_0(\boldsymbol{\sigma} \cdot \mathbf{p})}{m} + O(m^{-2}). \quad (4.2)$$

The antiparticle sector follows from the lower block by the hole-theory reinterpretation. An antiparticle of momentum $\mathbf{p}$ and spin $\mathbf{s}$ is the absence of a negative-energy solution of momentum $-\mathbf{p}$ and spin $-\mathbf{s}$, and its energy is minus the energy of that solution. A CPT test compares a particle with its CPT-conjugate state, since CPT takes a particle of momentum $\mathbf{p}$ and spin $\mathbf{s}$ to an antiparticle of the same momentum and spin $-\mathbf{s}$. Labelling that antiparticle state by the particle’s spin operator, its Hamiltonian is

$$H_{NR}^{\bar{f}}(\mathbf{p}, \boldsymbol{\sigma}) = -H_{\text{lower}}(-\mathbf{p}, \boldsymbol{\sigma}). \quad (4.3)$$

Since $\Sigma^i = \text{diag}(\sigma^i, \sigma^i)$ repeats the *same* operator σ^i in the upper and lower blocks,

$$\langle \text{upper} | b_i \gamma^0 \gamma_5 \gamma^i | \text{upper} \rangle = \langle \text{lower} | b_i \gamma^0 \gamma_5 \gamma^i | \text{lower} \rangle = -b_i \sigma^i, \quad (4.4)$$

and because b_i carries no momentum, Eq. (4.3) gives

$$H_{NR}^{\bar{f}}(b_i) = -(-\mathbf{b} \cdot \boldsymbol{\sigma}) = +\mathbf{b} \cdot \boldsymbol{\sigma}. \quad (4.5)$$

The CPT-conjugate antiparticle receives the opposite energy shift to the particle. This matter–antimatter sign flip is the signature of a CPT-odd coefficient. It is not a charge-conjugation effect: the axial-vector bilinear $\bar{\psi} \gamma_5 \gamma^\mu \psi$ is even under C for every μ , so C alone cannot produce the sign difference between Eqs. (4.1) and (4.5). The difference is a CPT effect, and it appears only when CPT-conjugate states are compared.

Stated at fixed physical spin, the result looks different. An antiparticle with the *same* spin as the particle has energy $-\mathbf{b} \cdot \mathbf{s}$, the same as the particle. This is Kostelecký and Lane’s antiparticle rule, which leaves b_μ , $c_{\mu\nu}$ and $g_{\lambda\mu\nu}$ unchanged and reverses a_μ , $d_{\mu\nu}$, e_μ , f_μ and $H_{\mu\nu}$ (Kostelecký and Lane, 1999b). The two statements are equivalent because a spin-linear term changes sign when the spin is reversed. For a CPT test the CPT-conjugate comparison is the relevant one: with it, and only with it, exact CPT symmetry gives $g_\alpha^{\bar{f}} = g_\alpha^f$ and $A_\alpha = 0$. We verified both comparisons symbolically for every coefficient family, using the full Foldy–Wouthuysen reduction and the kinetic-sector field redefinition. The CPT-conjugate comparison reproduces each coefficient’s CPT parity in every family, and the fixed-spin comparison reproduces Kostelecký and Lane’s rule in every family. Both comparisons require the momentum reversal in Eq. (4.3). Omitting it changes the result for every term that is odd in $\mathbf{p}$.¹ Promoting Eq. (4.1) to a two-body potential by

¹This agrees with Bluhm, Kostelecký and Russell’s (1999) treatment of hydrogen and antihydrogen. Their substitution rule takes $a_\mu \rightarrow -a_\mu$, $d_{\mu\nu} \rightarrow -d_{\mu\nu}$ and $H_{\mu\nu} \rightarrow -H_{\mu\nu}$ but leaves b_μ unchanged, which is the fixed-spin rule above. The b_μ -dependent shift in their 1S–2S frequency changes sign between the two atoms because “the hyperfine states in $\bar{\text{H}}$ have opposite positron and antiproton spin assignments relative to those of the electron and proton in H ”: the transitions being compared connect CPT-conjugate spin states.

pairing the same vertex on both fermions gives the spin–spin structure

$$V_2(r) = -g_A^{(1)} g_A^{(2)} \frac{1}{4\pi} (\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2) \frac{e^{-r/\lambda}}{r}, \quad (4.6)$$

identifying b_i with V_2 at leading order, while b_0 matches the velocity-dependent V_8 at order $1/m$.

Discussion: the asymmetry parameter does not saturate the way a naive substitution suggests. Substituting the exact, signed relation $g_{\bar{f}} = -g_f$ implied by Eqs. (4.1) and (4.5) directly into Eq. (2.22) gives

$$A_\alpha = \frac{g - (-g)}{g + (-g)} = \frac{2g}{0} \longrightarrow \infty, \quad (4.7)$$

a divergence, confirmed by a symbolic limit calculation, not a value that saturates at 1. This is not, however, what SPINDEP actually computes: its g_f and $g_{\bar{f}}$ inputs are independent, always-positive experimental *upper bounds*, not a signed measurement of one underlying coupling. For any two independent positive bounds g_1, g_2 , the ratio $(g_1 - g_2)/(g_1 + g_2)$ approaches ± 1 whenever one bound is far tighter than the other, *regardless of whether the underlying physics is CPT-odd or CPT-even*: substituting $g_{\text{tight}} = r g_{\text{loose}}$ and taking $r \rightarrow 0^+$ gives $A_\alpha \rightarrow 1$ with no CPT content at all. Consequently, an observed $|A_\alpha| \approx 1$ is *consistent with*, but not *evidence for*, a non-zero b_μ : the same numerical pattern is exactly what independent bounds of very different sensitivity would produce even under exact CPT symmetry. This result governs the interpretation of every asymmetry value reported later in this chapter.

4.1.2 The $H_{\mu\nu}$ Term

$H_{\mu\nu}$ is CPT-even but Lorentz-odd: CPT-conjugate matter and antimatter states receive identical shifts at tree level, so its experimental signature is an anisotropy (sidereal variation) rather than a matter–antimatter asymmetry. Its magnetic-like spatial components H_{ij} are even and generate, at leading order m^0 , a result obtained directly from $\gamma^0 \sigma^{ij}|_{\text{upper block}} = \varepsilon^{ijk} \sigma^k$ (verified by explicit 4×4 matrix multiplication for all three independent spatial pairs):

$$H_{NR}(H_{ij}) = +\boldsymbol{\mathcal{H}}_B \cdot \boldsymbol{\sigma}, \quad \mathcal{H}_B^k = \frac{1}{2} \varepsilon^{ijk} H_{ij}, \quad (4.8)$$

matching to the dipole–dipole potential V_3 . Its electric-like mixed components H_{0i} are odd, since $\gamma^0 \sigma^{0i} = i\gamma^i$ is block-off-diagonal, and generate, via the same FW cross-term

used for b_0 (Section 4.1.1) with the perturbing operator $\mathcal{O}_{H_E} = i \sum_k H_{0k} \gamma^k$, at order $1/m$,

$$H_{NR}(H_{0i}) = +\frac{1}{m} \boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{H}_E), \quad H_E^i \equiv H_{0i}, \quad (4.9)$$

matching to the spin-velocity potential V_7 . Applying Eq. (4.3) gives $H_{NR}^{\bar{f}} = H_{NR}^f$ for both components, as their CPT-even character requires. At fixed physical spin the sign reverses instead, in line with Kostelecký and Lane's rule $H_{\mu\nu} \rightarrow -H_{\mu\nu}$. This coefficient therefore gives $g_\alpha^{\bar{f}} = g_\alpha^f$ and $A_\alpha = 0$ for the underlying couplings, in contrast to b_μ .

4.1.3 The $d_{\mu\nu}$ Term

Following Kostelecký and Lane (1999a), $d_{\mu\nu}$ is a CPT-even, traceless, kinetic-sector coefficient entering through a derivative coupling, $\mathcal{L}_d = \frac{1}{2} i \bar{\psi} d^{\mu\nu} \gamma_5 \gamma_\mu \overleftrightarrow{\partial}_\nu \psi$. Direct computation gives $\Gamma_0 = -\gamma_5$ (odd) and $\Gamma_i = +\Sigma^i$ (even) for $\Gamma_\mu \equiv \gamma^0 \gamma_5 \gamma_\mu$, from which

$$H_{NR}(d_{i0}) = +d_{i0} m \sigma^i \quad (\text{order } m^1, \text{ mass-enhanced}) \rightarrow V_2, \quad (4.10)$$

$$H_{NR}(d_{ij}) = +d_{ij} p^j \sigma^i \quad (\text{order } m^0 \text{ in velocity}) \rightarrow V_7, V_8, \quad (4.11)$$

$$H_{NR}(d_{00}) = +d_{00} (\boldsymbol{\sigma} \cdot \mathbf{p}) \quad (\text{order } m^0 \text{ in momentum}) \rightarrow V_8. \quad (4.12)$$

All three results match Kostelecký and Lane's own non-relativistic Hamiltonian (1999a, Eq. 4) exactly, including the signs, once that equation is read in its own momentum convention. That convention is not stated explicitly in the text of Eq. 4, which describes p_j as the three-momentum of the particle, but it is fixed unambiguously by the free-particle term of the same authors' intermediate relativistic Hamiltonian (Kostelecký and Lane, 1999b), $m\mathcal{P}_0 := -p_j \gamma^0 \gamma^j$. That expression equals $-\boldsymbol{\alpha} \cdot \mathbf{p}$, so reproducing the required free Dirac Hamiltonian $+\boldsymbol{\alpha} \cdot \mathbf{p} + \beta m$ forces their lower-index p_j to be the covariant component $p_j = -p^j$ rather than the physical three-momentum. Read that way, the mass-enhanced d_{i0} term and both momentum-linear terms agree with Eq. 4 term by term; read as physical momentum, the two momentum-linear terms appear to differ by an overall sign while the mass term still agrees, which is the signature of a momentum-index mismatch rather than of a derivation error. The full comparison, together with an independent implementation of Kostelecký and Lane's (1999b) field-redefinition procedure, is carried out symbolically in the supporting notebooks.

4.1.4 The $g_{\lambda\mu\nu}$ Term

$g_{\lambda\mu\nu}$ is CPT-odd and dimensionless, and enters the kinetic sector through $\frac{1}{2} g_{\lambda\mu\nu} \sigma^{\lambda\mu}$ in Γ_ν . It is antisymmetric in its first two indices, since $\sigma^{\lambda\mu}$ is; the third is the derivative index. Only the $\nu = 0$ slice enters Γ_0 , so the same kinetic-sector field redefinition used for $d_{\mu\nu}$ applies, and the resulting Hamiltonian is Hermitian.

Writing $\tilde{g}_j \equiv \frac{1}{2}\varepsilon_{jkl}g_{kl0}$ for the dual of the spatial–spatial $\nu = 0$ block and $(\mathbf{g}_{00})_k \equiv g_{k00}$, the reduction separates by index family:

$$H_{NR}(g_{[kl]0}) = -m \tilde{\mathbf{g}} \cdot \boldsymbol{\sigma} \quad (\text{order } m^1, \text{ mass-enhanced}) \rightarrow V_2, \quad (4.13)$$

$$H_{NR}(g_{[0k]0}) = +\boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{g}_{00}) \quad (\text{order } m^0) \rightarrow V_7, \quad (4.14)$$

$$H_{NR}(g_{[kl]j}) = +\frac{1}{2}\varepsilon_{klm}g_{mlj} p^j \sigma^k \quad (\text{order } m^0 \text{ in velocity}) \rightarrow V_7, V_8, \quad (4.15)$$

with $g_{[0k]j}$ contributing nothing at this order. All three reproduce Kostecký and Lane’s (1999a, Eq. 4) non-relativistic Hamiltonian term by term, under the same covariant momentum convention established for $d_{\mu\nu}$ in Section 4.1. Since that convention was fixed there without reference to $g_{\lambda\mu\nu}$, its also making this sector agree is an independent confirmation of the reading.

Matter–antimatter behaviour. Applying Eq. (4.3) to each family gives an exact sign flip for all three contributing families, as the CPT-odd character of $g_{\lambda\mu\nu}$ requires. The momentum reversal in that equation is essential here. The $g_{[0k]0}$ and $g_{[kl]j}$ terms are linear in $\mathbf{p}$, and comparing the lower block at the same momentum instead would wrongly suggest that these two families do not flip. At fixed physical spin, all three families give identical particle and antiparticle shifts, in line with Kostecký and Lane’s rule $g_{\lambda\mu\nu} \rightarrow +g_{\lambda\mu\nu}$. The matter–antimatter signature is therefore set by the coefficient’s CPT parity, for $g_{\lambda\mu\nu}$ as for b_μ .

4.1.5 Master SME→DM Matching Table

Table 4.1 synthesises the four derivations above. Its final column gives the relation between the particle and antiparticle couplings for CPT-conjugate states. That relation describes the underlying couplings, not the published bounds: as Section 4.1.1 shows, an A_α computed from one-sided bounds cannot reveal it.

4.2 The Compiled Constraint Database

Applying the compilation methodology of Section 3.2.1 produced a database of 283 experimental constraint datasets, of which 247 enter the physics analysis. Thirty-six are compiled into the registry but held out of it, for five separately stated reasons. Twenty-eight are **Combined** envelope curves, twelve in $g_A g_V$ and sixteen in $g_p g_s$: aggregates over several experiments, which the source database labels **combined** and marks **review_only**, so that no single V_n applies to them by construction. One is a $g_p g_p$ astrophysical e - N bound that carries no potential assignment and for which none can be established from the source database. Two are **Code-plot-matlab** contributor-template placeholders, whose coupling

SME coeff.	CPT	NR Hamiltonian	DM potential(s)	1/m order	Matter– antimatter [†]
b_i	Odd	$-\mathbf{b} \cdot \boldsymbol{\sigma}$ (matter), $+\mathbf{b} \cdot \boldsymbol{\sigma}$ (CPT-conjugate antimatter)	V_2	m^0	$g^{\bar{f}} = -g^f$
b_0	Odd	$-b_0(\boldsymbol{\sigma} \cdot \mathbf{p})/m$	V_8	m^{-1}	$g^{\bar{f}} = -g^f$
H_{ij}	Even	$+\mathcal{H}_B \cdot \boldsymbol{\sigma}$	V_3	m^0	$g^{\bar{f}} = g^f$
H_{0i}	Even	$+\frac{1}{m}\boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{H}_E)$	V_7	m^{-1}	$g^{\bar{f}} = g^f$
d_{i0}	Even	$+d_{i0} m \sigma^i$	V_2	m^1 (!)	$g^{\bar{f}} = g^f$
d_{ij}	Even	$+d_{ij} p^j \sigma^i$	V_7, V_8	m^0 (vel.)	$g^{\bar{f}} = g^f$
d_{00}	Even	$+d_{00}(\boldsymbol{\sigma} \cdot \mathbf{p})$	V_8	m^0 (mom.)	$g^{\bar{f}} = g^f$
$g_{[kl]0}$	Odd	$-m \tilde{\mathbf{g}} \cdot \boldsymbol{\sigma}$	V_2	m^1	$g^{\bar{f}} = -g^f$
$g_{[0k]0}$	Odd	$+\boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{g}_{00})$	V_7	m^0	$g^{\bar{f}} = -g^f$
$g_{[kl]j}$	Odd	$+\frac{1}{2}\varepsilon_{klm}g_{mlj}p^j\sigma^k$	V_7, V_8	m^0 (vel.)	$g^{\bar{f}} = -g^f$

Table 4.1: Master SME→DM matching table, covering the complete set of SME coefficients that generate spin-dependent non-relativistic potentials. Here $\tilde{g}_j \equiv \frac{1}{2}\varepsilon_{jkl}g_{kl0}$ and $(\mathbf{g}_{00})_k \equiv g_{k00}$. [†]Relation between the particle and antiparticle couplings for CPT-conjugate states (antiparticle spin reversed), verified symbolically for every row; it follows the coefficient’s CPT parity without exception. At fixed physical spin every sign in this column reverses (Kostelecký and Lane, 1999b). Neither relation gives $|A_\alpha| \rightarrow 1$: a flip makes the signed A_α diverge (Eq. 4.7), and equal couplings give $A_\alpha = 0$. A value near 1 in the data therefore reflects a sensitivity gap between independent bounds.

label is a source directory named for its file format rather than its physics content. Five are the 90% confidence-level variants of the Cong *et al.* (2025) hydrogen-spectroscopy bounds, which the source database ships alongside the 95% CL curves; the two sets are the same measurement reported at two confidence levels, so counting both would double-count the experiment and pair each twice against the same antimatter curve. The 95% CL set is retained as the more conservative limit and the only one the source database carries a curation record for. Table 4.2 gives both counts by coupling family; every count reported in the remainder of this chapter is an analysed count unless stated otherwise.

By DM potential, the analysed database is dominated by V_{4+5} (61 datasets), followed by V_1 (47), V_3 (39), V_{9+10} (31), V_2 (20), and V_{12+13} (16), with V_{1a} (14), V_{11} (7), V_8 (6), V_{2+3} (2), V_{16} (2), and V_{15} (2) making up the remainder. A bare numeric prefix in the source filenames denotes the potential number itself, so that a 3-prefixed file is a bound on V_3 alone and only the 23 prefix denotes the $V_2 + V_3$ combination; every assignment in this table has been validated against the source database’s own curation records, with no disagreement among the 194 datasets those records cover. Two successive filename-convention passes resolved 26 of the 39 files that initially carried no recoverable potential token, by giving each an explicit potential-number prefix the parser could read; the 13 that remained are among the 36 datasets held out of the analysis above. Of the 247

Coupling family	Compiled	Analysed
$g_A g_A$	69	66
$g_p g_s$	53	37
$g_s g_s$	49	49
$g_V g_V$	45	44
$g_A g_V$	37	25
$g_p g_p$	17	15
V_1 (unlabelled coupling)	11	11
Code-plot-matlab (template placeholder)	2	0
Total	283	247

Table 4.2: Dataset count by coupling family, as compiled into the registry and as admitted to the physics analysis.

analysed datasets, only 22 involve an antimatter sector: 9 in $e\bar{p}$, 6 in $e\bar{\mu}$, 5 in $e-e^+$, and one each in $\bar{p}$ -He and the $dd\mu^+$ molecular ion (both, per Cong *et al.*, 2025, laboratory bounds on $\bar{p}$ - N and μ^+ - N respectively). The six $e\bar{\mu}$ bounds all come from muonium, which contains an antimuon, so they belong on the antimatter side for the same reason $dd\mu^+$ does. This imbalance, 22 antimatter datasets against 225 matter-sector ones, is the central empirical fact the remainder of this chapter works with.

4.3 Matter–Antimatter Asymmetry Analysis

4.3.1 The Fifteen Matched Pairs

SPINDEP’s pairing logic (Section 3.2.1) matched fifteen matter–antimatter pairs from the analysed database. Table 4.3 gives, for each, the mean absolute asymmetry, the autocorrelation-corrected effective degrees of freedom, the resulting significance, and the 95% bootstrap confidence interval.

Effective degrees of freedom range from 6 to 21 across the fifteen pairs, a factor of 14–50 below the 300 grid points, so the autocorrelation correction of Section 3.2.2 is far from cosmetic. Every pair stays significant after it. The weakest is the Jiao/Karshenboim $e-e^+$ pair at $Z_{\text{eff}} = 5.1$ ($p \approx 2 \times 10^{-7}$); the others lie between $Z_{\text{eff}} = 26$ and 51.

Two rows repeat two others. The source compilation files the same Fadeev *et al.* (2022) V_3 curves under both $g_V g_V$ and $g_p g_p$, so the $g_V g_V$ and $g_p g_p$ pairs built from them give identical numbers. The fifteen pairs therefore contain thirteen independent comparisons.

4.3.2 Discussion

Per Section 4.1.1, an asymmetry value near 1 in Table 4.3 is *consistent with*, but not *evidence for*, CPT violation: a large sensitivity gap between the matter- and antimatter-

Coupling	Potential	Sector	Matter / antimatter source	$\overline{ A_\alpha }$	dof _{eff}	Z _{eff}	95% CI
$g_A g_A$	V_3	$e\text{-}\bar{p}$	Cong <i>et al.</i> (2025) / Fadeev <i>et al.</i> (2022)	1.0000	8	51.1	[1.0000, 1.0000]
$g_A g_A$	V_2	$e\text{-}\bar{p}$	Cong <i>et al.</i> (2025) / Ficek <i>et al.</i> (2018)	0.9999	21	34.4	[0.9999, 0.9999]
$g_A g_A$	V_2	$e\text{-}\bar{p}$	Karshenboim (2011) / Ficek <i>et al.</i> (2018)	0.9998	17	38.8	[0.9998, 0.9998]
$g_A g_A$	V_2	$e\text{-}e^+$	Ficek <i>et al.</i> (2017) / Karshenboim (2011)	0.9892	21	48.2	[0.9886, 0.9896]
$g_A g_A$	V_3	$e\text{-}e^+$	Ficek <i>et al.</i> (2017) / Fadeev <i>et al.</i> (2022)	0.9539	6	38.5	[0.9518, 0.9557]
$g_V g_V$	V_3	$e\text{-}e^+$	Fadeev <i>et al.</i> (2022) / Fadeev <i>et al.</i> (2022)	0.9535	6	37.1	[0.9516, 0.9552]
$g_p g_p$	V_3	$e\text{-}e^+$	Fadeev <i>et al.</i> (2022) / Fadeev <i>et al.</i> (2022)	0.9535	6	37.1	[0.9516, 0.9552]
$g_p g_p$	V_3	$e\text{-}\bar{p}$	Cong <i>et al.</i> (2025) / Ficek <i>et al.</i> (2018)	0.9304	6	34.3	[0.9293, 0.9314]
$g_A g_A$	V_3	$e\text{-}\bar{p}$	Cong <i>et al.</i> (2025) / Ficek <i>et al.</i> (2018)	0.9303	6	34.3	[0.9292, 0.9313]
$g_s g_s$	V_1	$e\text{-}e^+$	Delaunay <i>et al.</i> (2017) / Adkins <i>et al.</i> (2022)	0.8727	21	25.7	[0.8686, 0.8770]
$g_A g_A$	V_3	$e\text{-}\bar{p}$	Fadeev <i>et al.</i> (2022) / Ficek <i>et al.</i> (2018)	0.8236	6	37.3	[0.8168, 0.8299]
$g_A g_A$	V_3	$e\text{-}\bar{p}$	Fadeev <i>et al.</i> (2022) / Fadeev <i>et al.</i> (2022)	0.8044	8	43.8	[0.8024, 0.8063]
$g_V g_V$	V_3	$e\text{-}\bar{p}$	Fadeev <i>et al.</i> (2022) / Ficek <i>et al.</i> (2018)	0.7994	7	31.8	[0.7971, 0.8015]
$g_p g_p$	V_3	$e\text{-}\bar{p}$	Fadeev <i>et al.</i> (2022) / Ficek <i>et al.</i> (2018)	0.7994	7	31.8	[0.7971, 0.8015]
$g_A g_A$	V_2	$e\text{-}e^+$	Jiao <i>et al.</i> (2020) / Karshenboim (2011)	0.3336	9	5.1	[0.3197, 0.3474]

Table 4.3: All fifteen matched matter–antimatter pairs, with the autocorrelation-corrected effective degrees of freedom, the weighted- χ^2 significance Z_{eff} after that correction (one-sided Gaussian equivalent), and the 95% bootstrap confidence interval on the mean asymmetry.

sector measurement being compared produces exactly the same signature regardless of the true underlying physics. This applies uniformly to every row, including the $g_{AgA} \cdot V_3 \cdot e\text{-}\bar{p}$ pair at 1.0000, the highest asymmetry in the database. The $g_s g_s \cdot V_1 \cdot e\text{-}e^+$ pair at 0.8727 makes the same point from a different direction. V_1 is spin-independent, so none of the four coefficients in Table 4.1 feeds it (only a_μ , $c_{\mu\nu}$ and e_μ enter spin-independent terms), yet its asymmetry sits well inside the range spanned by the g_{AgA} pairs, 0.33 to 1.00. A high asymmetry is simply what two bounds of very different sensitivity produce, whatever channel they sit in. The lowest asymmetry in the table, 0.334 for the $g_{AgA} \cdot V_2 \cdot e\text{-}e^+$ pair (Jiao *et al.*, 2020, versus Karshenboim, 2011), is the pair whose matter- and antimatter-sector bounds are closest in sensitivity among the fifteen, and is therefore the single most informative data point currently in the database for constraining b_μ in the $e\text{-}e^+$ channel, precisely because it is *not* saturated near $|A_\alpha| \approx 1$.

The four pairs built on the Cong *et al.* (2025) hydrogen bounds show the effect directly. Each can be set against a pair with the same coupling, potential and antimatter partner but an older $e\text{-}p$ bound on the matter side, and in all four the newer bound gives the higher asymmetry: 0.9999 against 0.9998 for $g_{AgA} \cdot V_2$ with Ficek *et al.* (2018), 1.0000 against 0.8044 for $g_{AgA} \cdot V_3$ with Fadeev *et al.* (2022), and 0.9303 against 0.8236 (g_{AgA}) and 0.9304 against 0.7994 ($g_p g_p$) for V_3 with Ficek *et al.* (2018). Improving the matter-sector measurement, with no change whatever to the antimatter sector or to the underlying physics, drove the asymmetry *up*. This is the clearest available demonstration that A_α built from one-sided bounds tracks the sensitivity ratio rather than CPT status, and it is a direct empirical confirmation of the analytic argument of Section 4.1.1.

4.3.3 Figures

Figures 4.1–4.13 show the constraint atlas for each of the twelve classified potential labels in the compiled database. V_1 carries two panels: the bounds catalogued as a coupling $|g|$ (Figure 4.1) and those catalogued as the gravity-normalised Yukawa strength $|\alpha|$ (Figure 4.2), plotted separately because the two conventions typically differ by more than thirty orders of magnitude and cannot share a y-axis. Several labels, V_{2+3} , V_{4+5} , V_{9+10} , and V_{12+13} , each combine two adjacent catalogue potentials that the underlying experiments do not distinguish individually. V_{2+3} is the sparsest of these, with only two datasets: a bound is filed under it only where the source explicitly reports the $V_2 + V_3$ combination, while a bound on V_3 alone is filed under V_3 . V_6 , V_7 , and V_{14} have no classified dataset at all. Figures 4.14–4.15 show the matter–antimatter comparison for the highest- and lowest-asymmetry pairs in Table 4.3, and Figure 4.16 shows the V_1 pair, included separately since V_1 is, after the filename-convention pass discussed in Section 4.2, now the second most populated potential in the database.

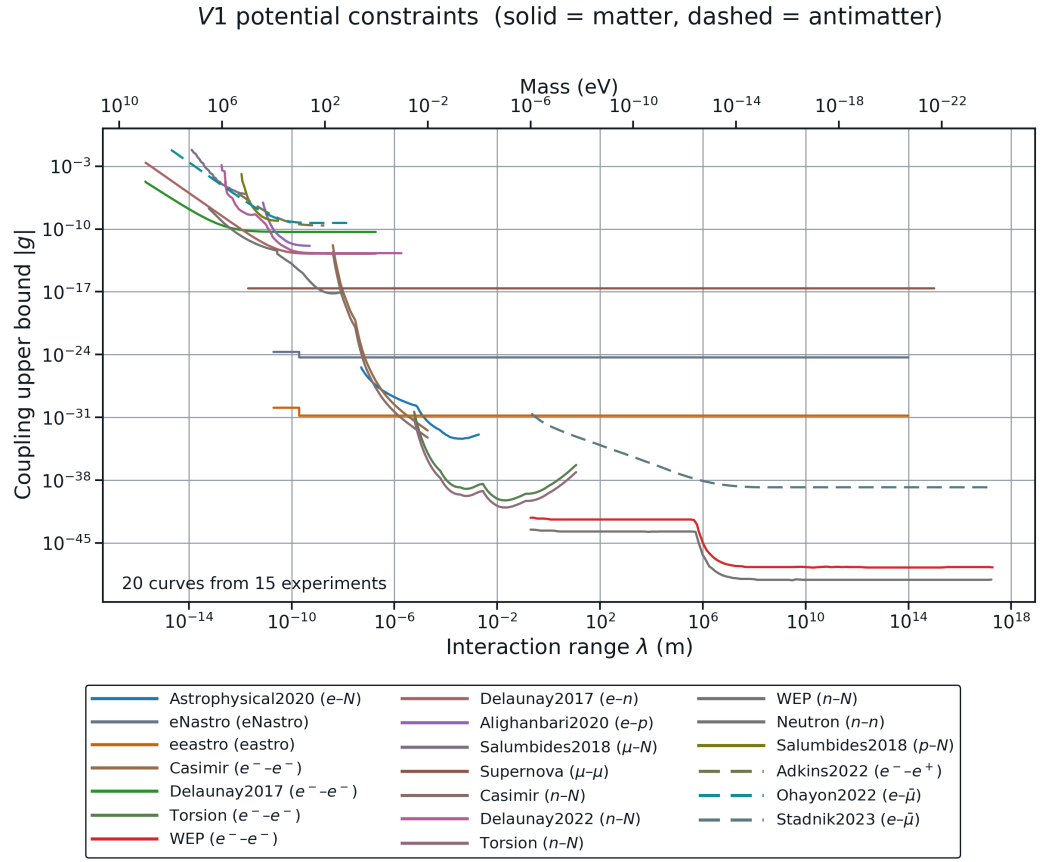


Figure 4.1: Constraint atlas for V_1 .

V_1 potential constraints — gravity-normalised α (solid = matter, dashed = antimatter)

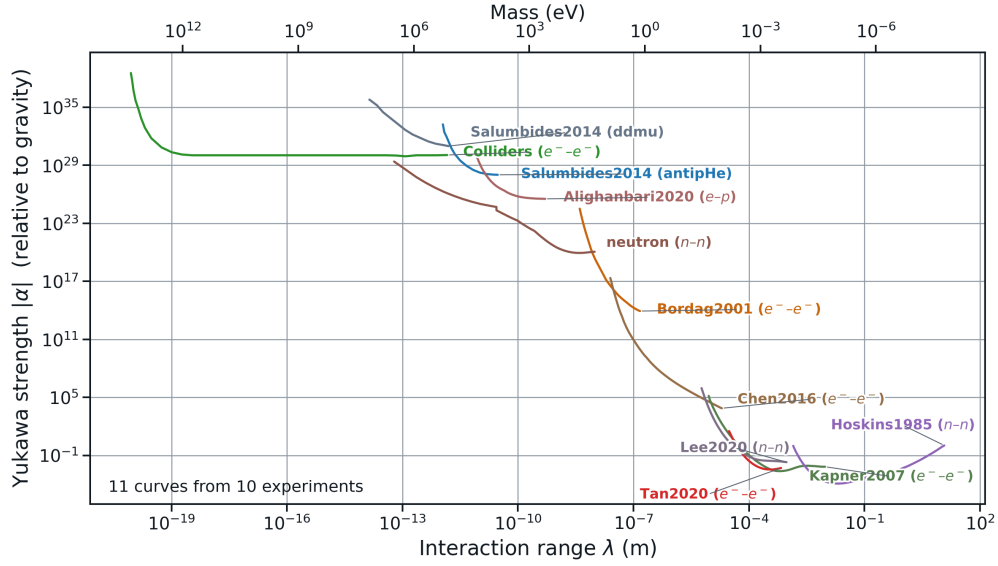


Figure 4.2: Constraint atlas for V_1 , gravity-normalised. These bounds are published as the Yukawa strength $|\alpha|$ relative to gravity rather than as a coupling $|g|$, and are shown on a separate axis because the two conventions typically differ by more than thirty orders of magnitude.

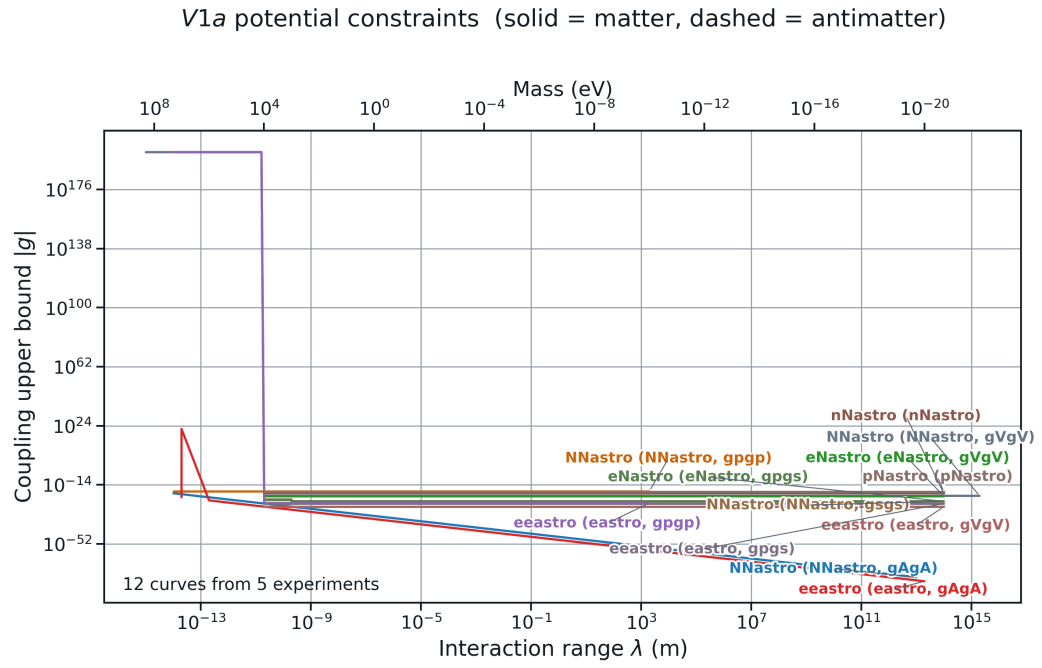


Figure 4.3: Constraint atlas for V_{1a} .

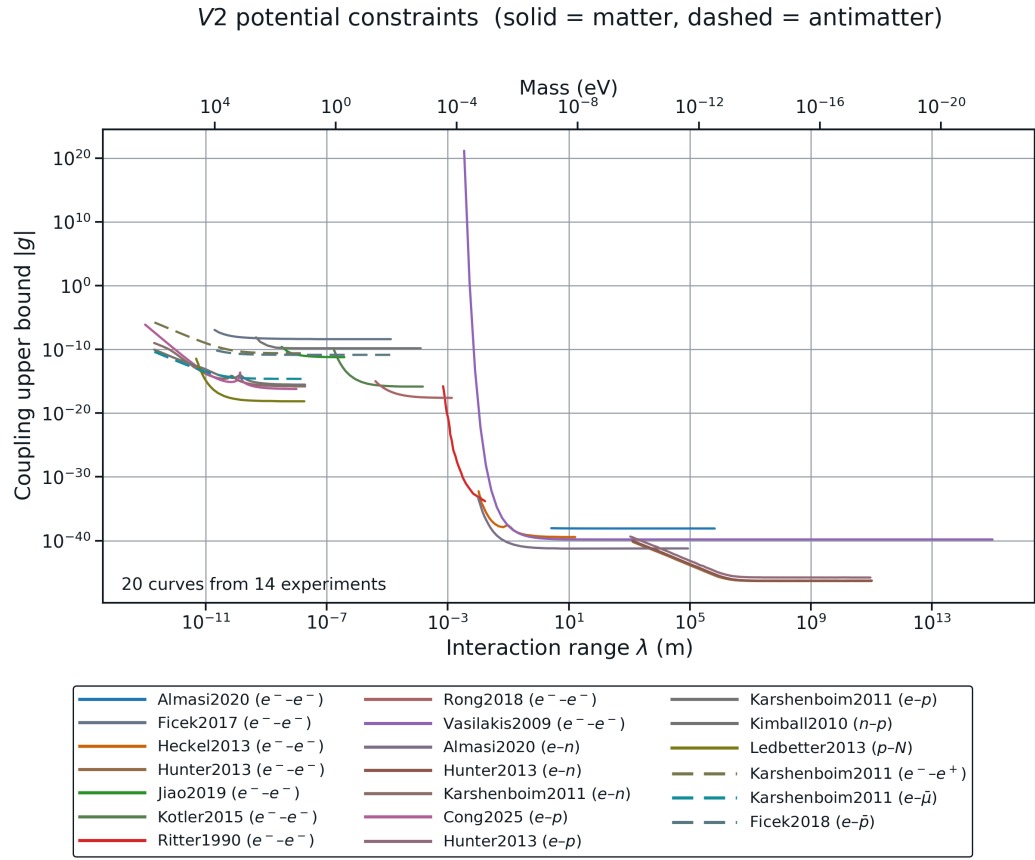


Figure 4.4: Constraint atlas for V_2 .

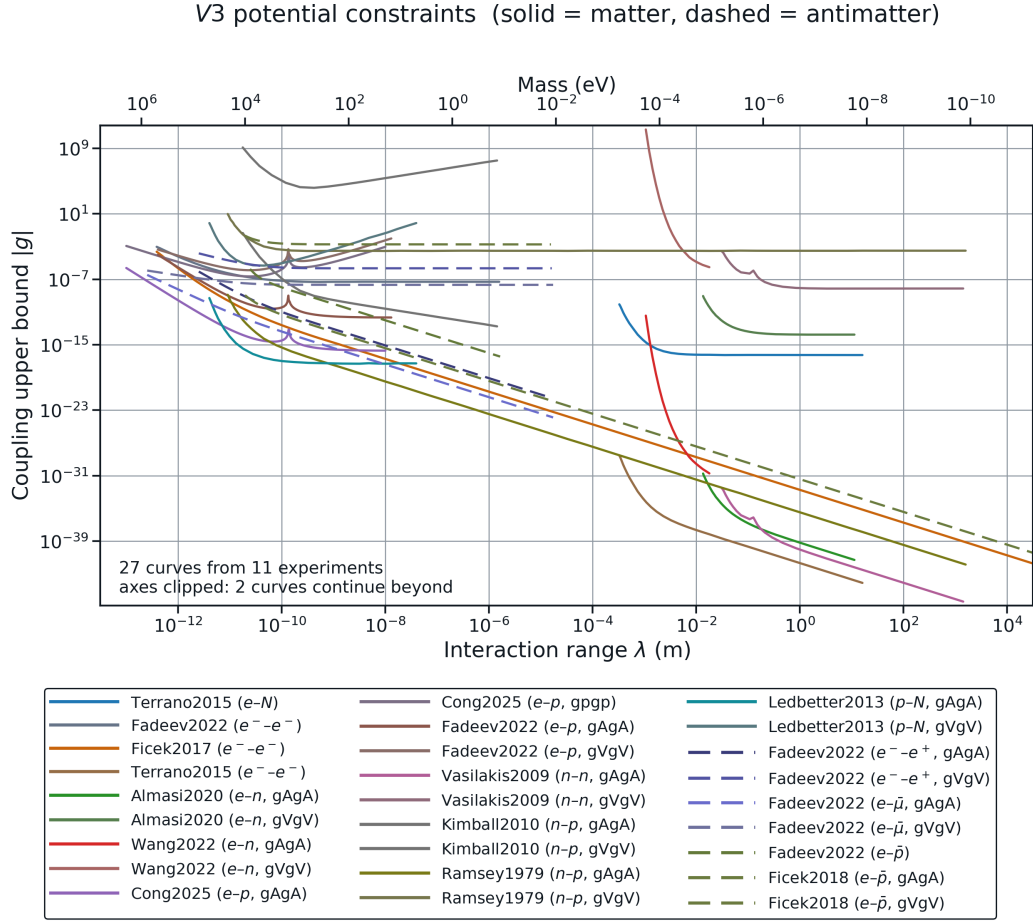


Figure 4.5: Constraint atlas for V_3 , the best-covered potential in the antimatter sector (39 datasets, 10 of them antimatter).

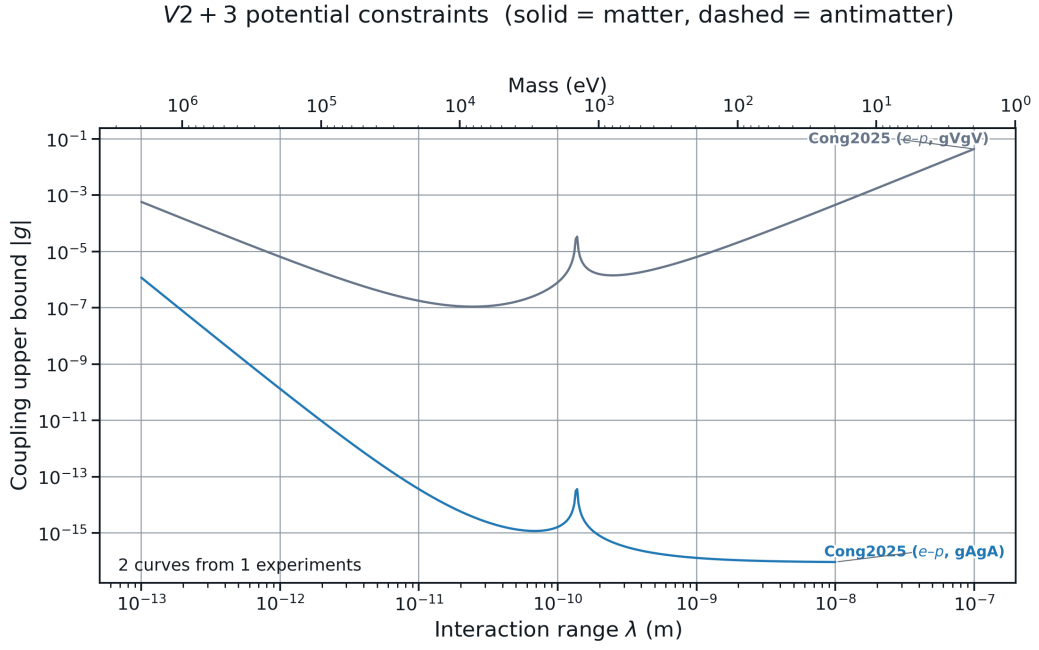


Figure 4.6: Constraint atlas for V_{2+3} . Only two datasets carry this combination label, both matter-sector; the great majority of what earlier drafts filed under V_{2+3} are V_3 -only bounds and appear in Figure 4.5.

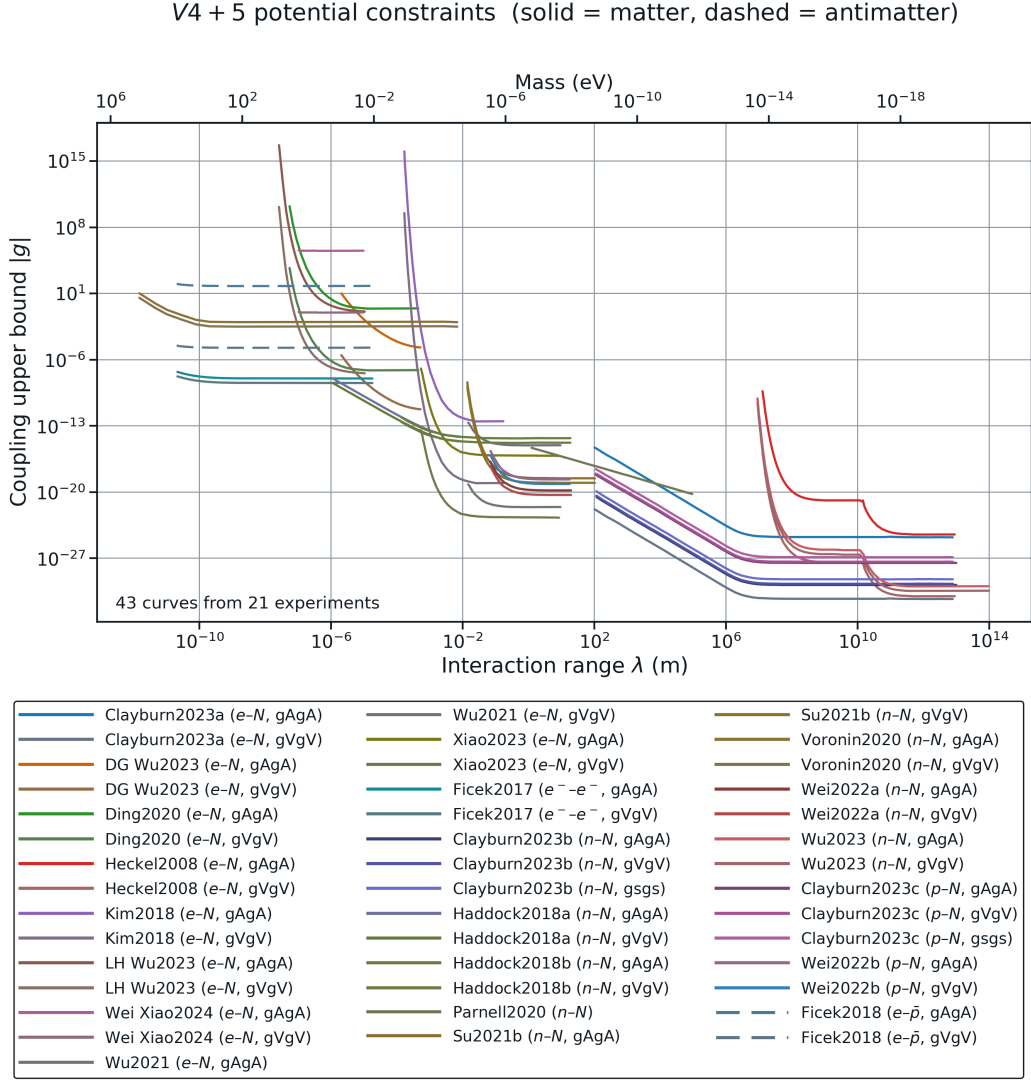


Figure 4.7: Constraint atlas for V_{4+5} , the most populated potential in the database (61 datasets).

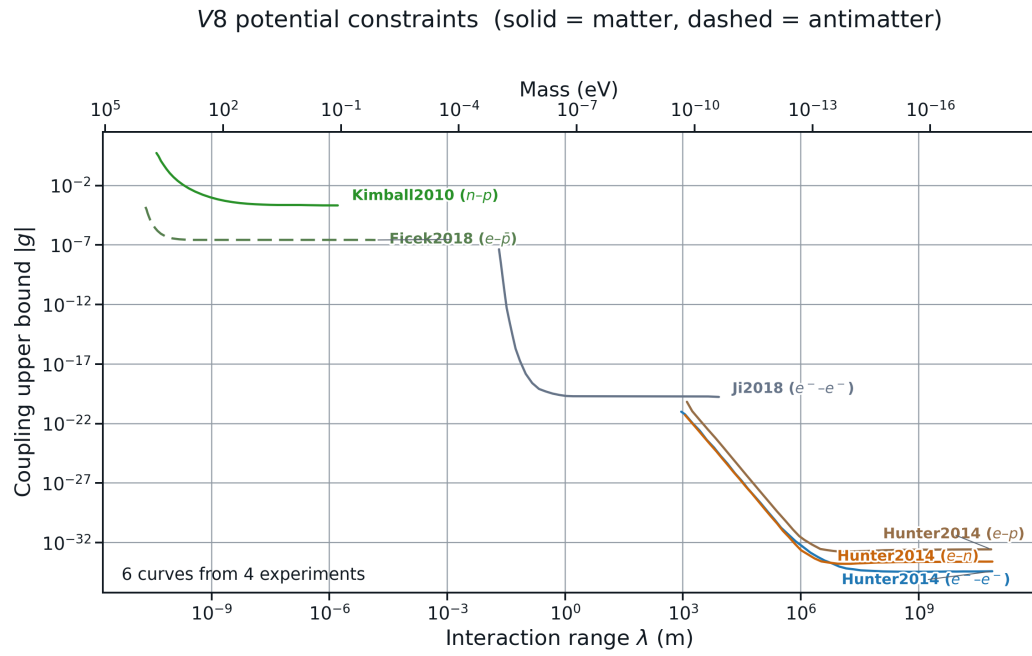


Figure 4.8: Constraint atlas for V_8 .

$V_9 + 10$ potential constraints (solid = matter, dashed = antimatter)

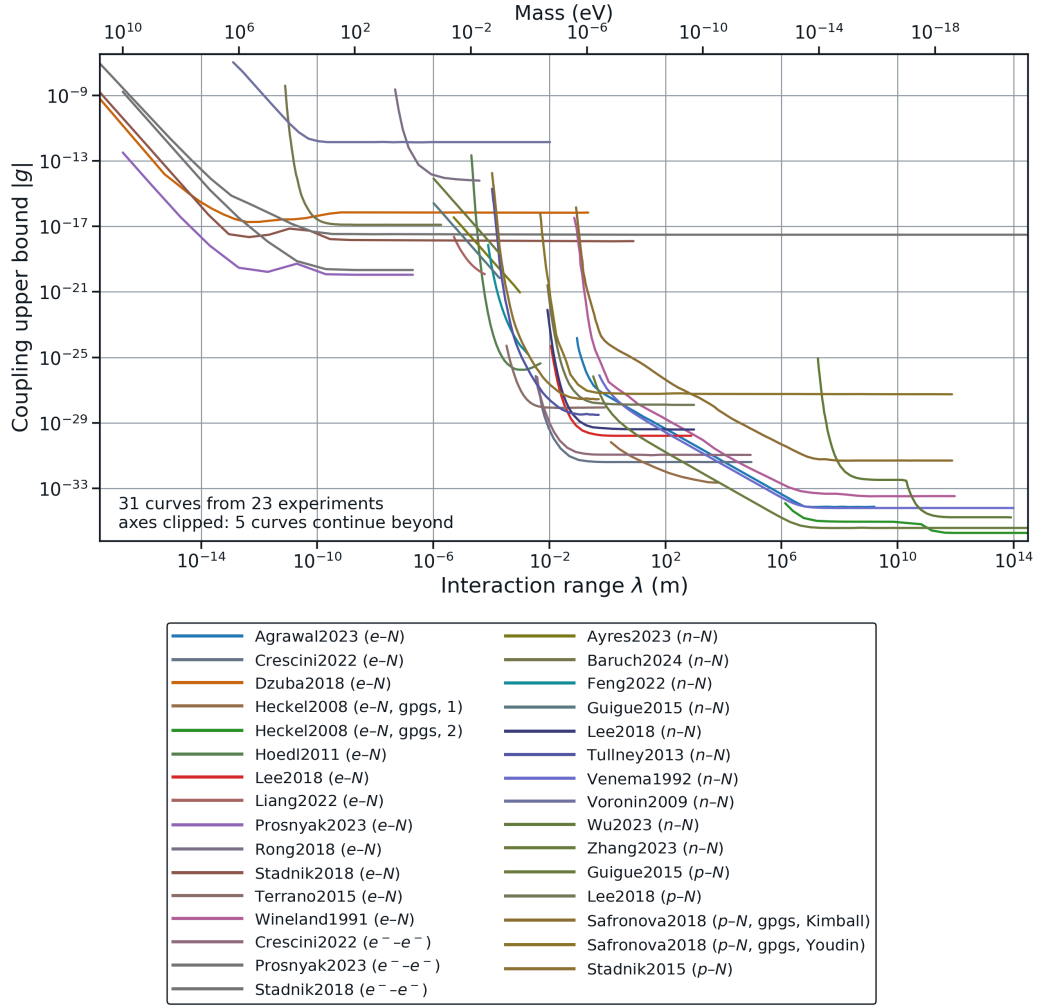


Figure 4.9: Constraint atlas for V_{9+10} .

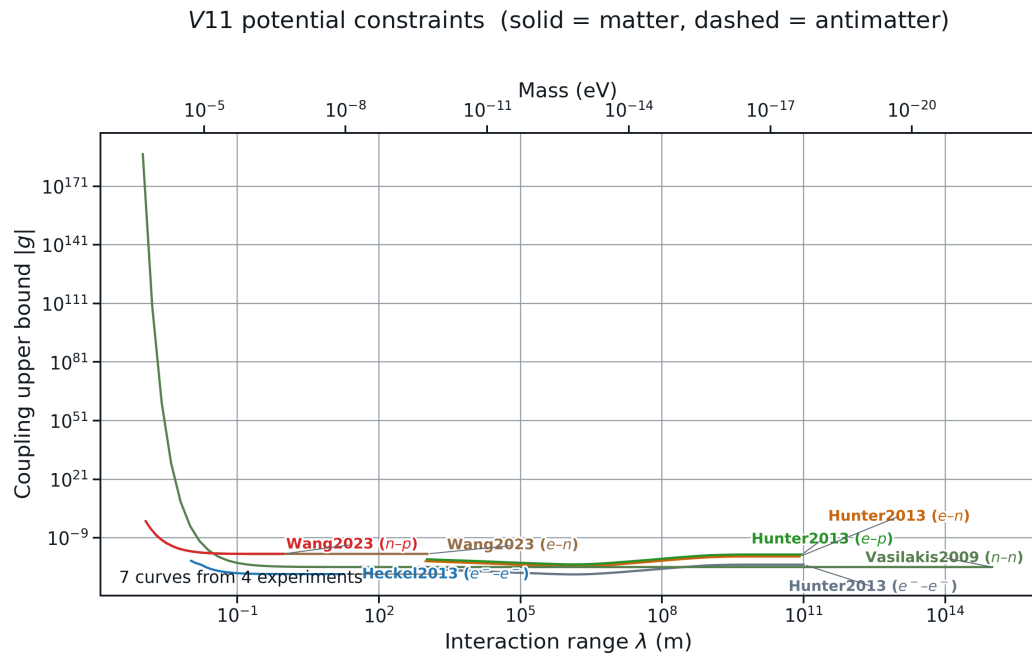


Figure 4.10: Constraint atlas for V_{11} .

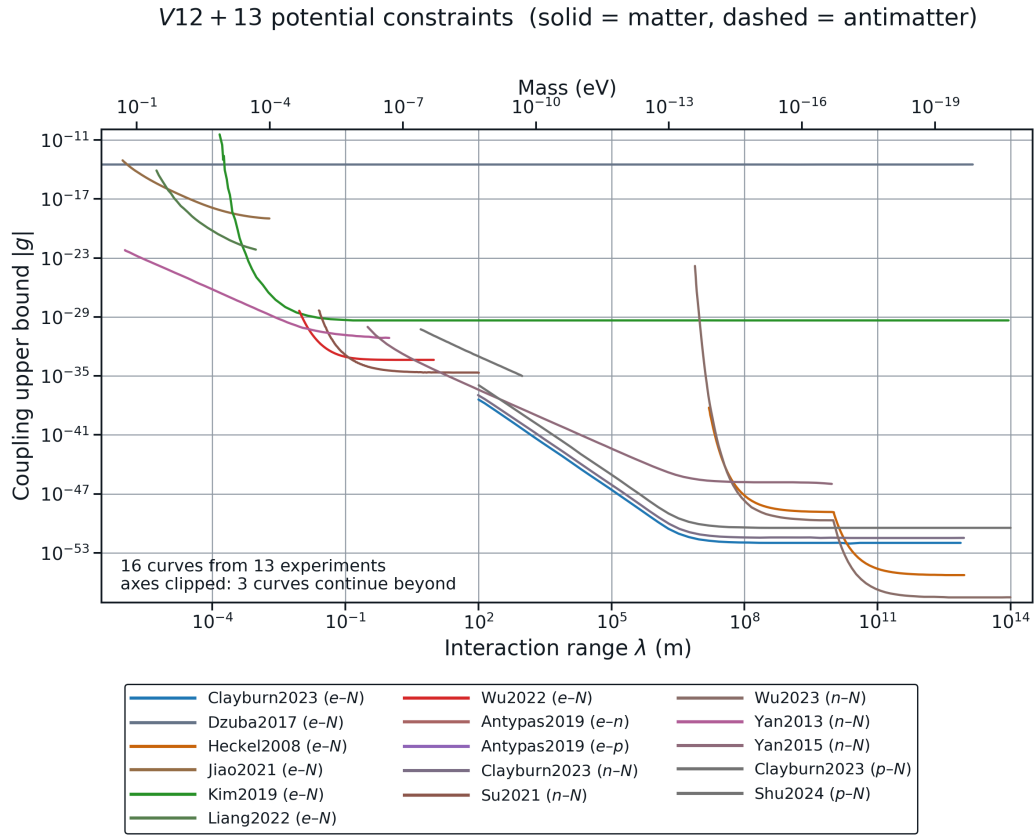


Figure 4.11: Constraint atlas for V_{12+13} .

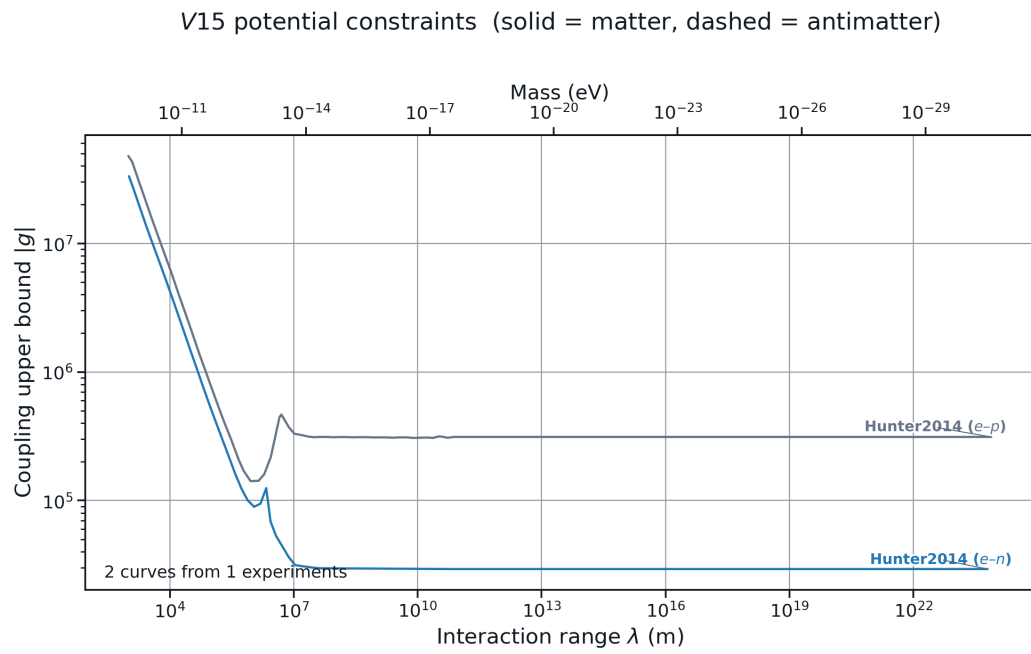


Figure 4.12: Constraint atlas for V_{15} .

V_{16} potential constraints (solid = matter, dashed = antimatter)

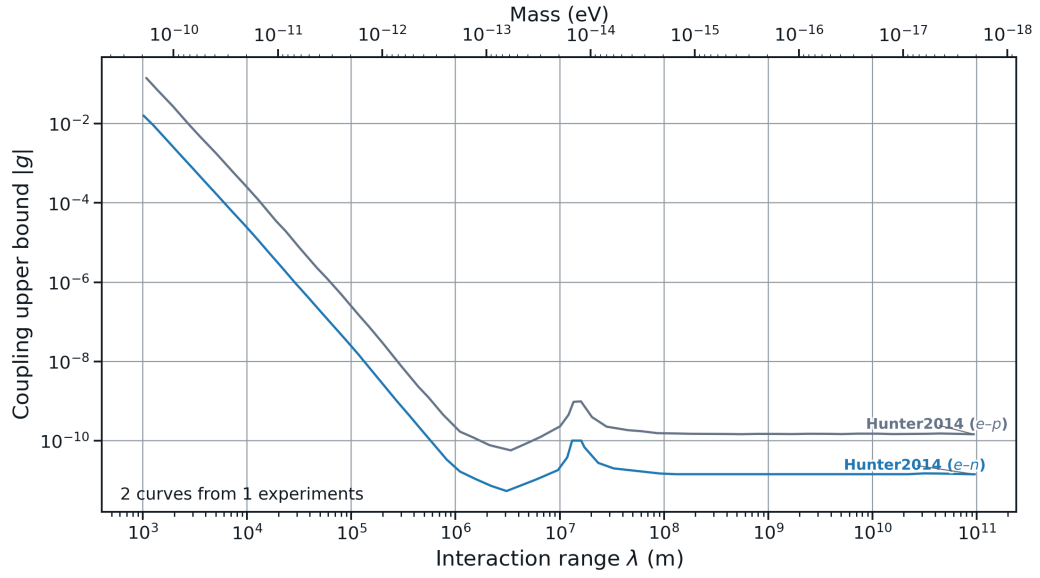


Figure 4.13: Constraint atlas for V_{16} .

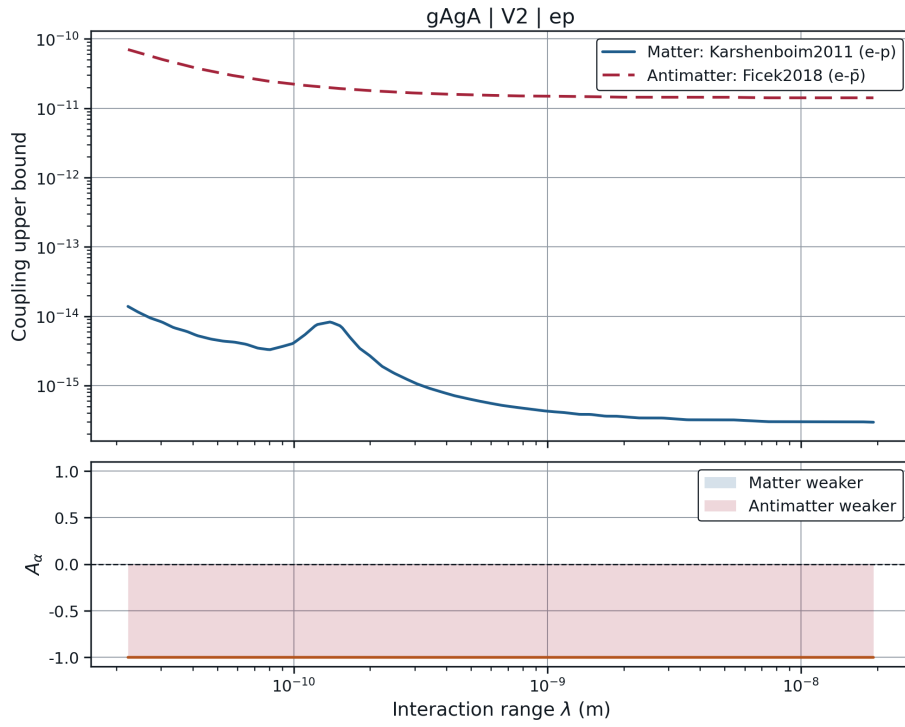


Figure 4.14: Matter–antimatter comparison for the $g_{AgA} \cdot V_2 \cdot e\text{-}\bar{p}$ pair (Karshenboim, 2011, versus Ficek *et al.*, 2018), one of the three highest asymmetries in the compiled database ($|\overline{A_\alpha}| = 0.9998$).

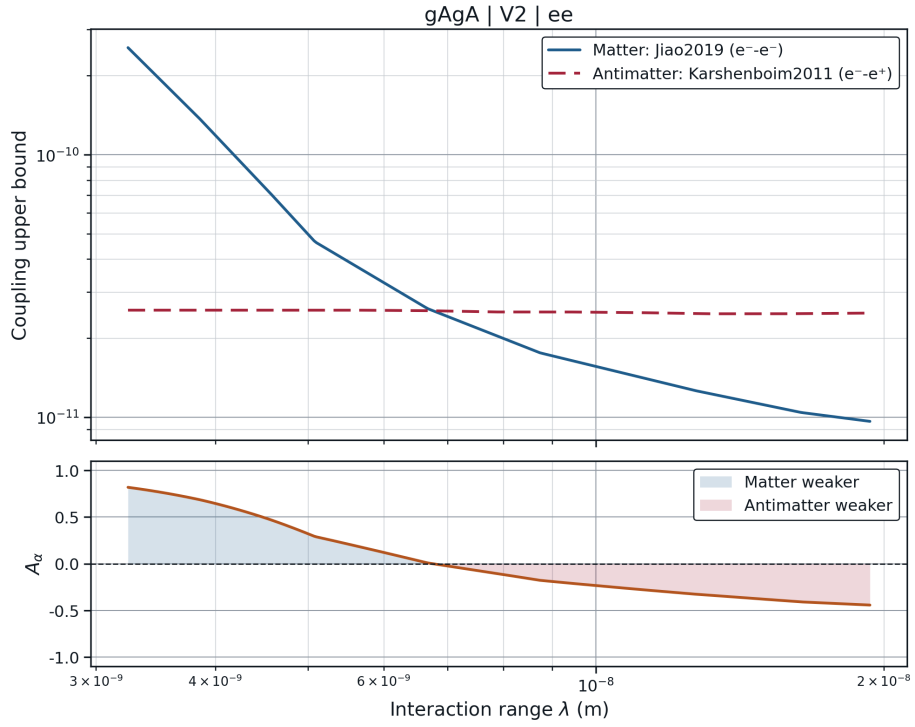


Figure 4.15: Matter–antimatter comparison for the $g_{AgA} \cdot V_2 \cdot e-e^+$ pair (Jiao *et al.*, versus Karshenboim, 2011), the lowest asymmetry in the compiled database ($|A_\alpha| = 0.334$), included alongside Figure 4.14 to show that the analysis does not produce a saturated asymmetry for every pair by construction.

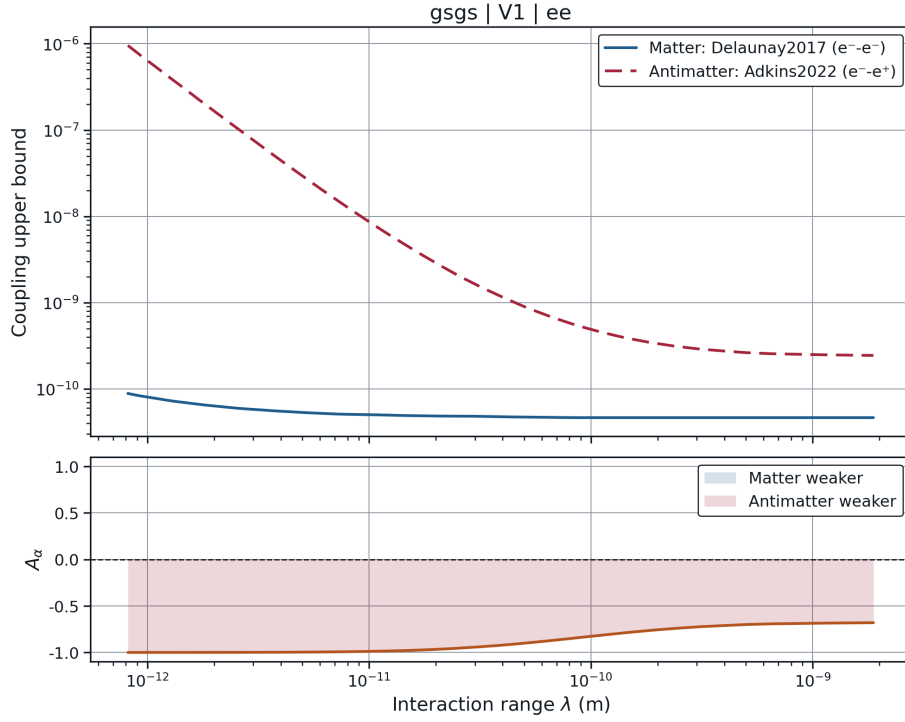


Figure 4.16: Matter–antimatter comparison for the $g_s g_s \cdot V_1 \cdot e^-e^+$ pair (Delaunay *et al.*, 2017, versus Adkins *et al.*, 2022), the only currently matched V_1 pair. V_1 is spin-independent, so none of the four spin-dependent coefficients feeds it, yet its asymmetry is comparable to that of the spin-dependent pairs (see the discussion following Table 4.3).

4.4 Gap Analysis

4.4.1 Coverage by Fermion Sector

Table 4.4 compares the number of analysed datasets on each side of the eleven canonical matter–antimatter *sector* pairs used by SPINDEP. These are pairings of fermion sectors, not the matched dataset pairs of Section 4.3.

Matter sector	Antimatter sector	Matter	Antimatter	Status
$e-e$	$e-e^+$	40	5	Populated (used above)
$e-p$	$e-\bar{p}$	17	9	Populated (used above)
$e-\mu$	$e-\bar{\mu}$	0	6	Antimatter only (muonium)
$e-n$	$e-\bar{n}$	16	0	Gap
$\mu-\mu$	$\mu-\bar{\mu}$	1	0	Gap (thin on both sides)
$n-p$	$n-\bar{p}$	9	0	Gap
$n-n$	$n-\bar{n}$	8	0	Gap
$p-p$	$p-\bar{p}$	0	0	No data either side
$e-N$	$e-\bar{N}$	50	0	Largest gap
$n-N$	$n-\bar{N}$	49	0	Gap (second largest)
$p-N$	$p-\bar{N}$	18	0 [†]	Gap in comparable units

Table 4.4: Matter versus antimatter dataset counts per canonical fermion sector pair. [†]A $\bar{p}-N$ bound does exist, from the antiprotonic-helium spectroscopy analysed by Salumbides *et al.* (2014); it is excluded from this count because it is published as a gravity-normalised strength $|\alpha|$ rather than a coupling bound $|g|$ and so cannot be compared with the $p-N$ datasets without a conversion this framework does not perform. The gap in *comparable* data is therefore real, but it is a unit-conversion gap rather than an absence of measurement.

The three nucleus sectors dominate the gap. $e-N$ is the single largest opportunity in the database, with 50 matter-sector datasets against zero in $e-\bar{N}$, but $n-N$ is barely behind it at 49, and $p-N$ adds a further 18; between them these three account for 117 of the 225 matter-sector datasets and carry no antimatter-sector counterpart at all. The $e-n$, $n-p$, and $n-n$ sectors each have a smaller but still substantial matter-sector base (8–16 datasets), again with no antimatter-sector counterpart. The $e-\mu$ row is the mirror image: all six of its bounds come from muonium and so sit on the antimatter side, with no matter-sector $e-\mu$ bound to compare them with.

The eleven sector pairs in Table 4.4 account for 208 matter-sector and 20 antimatter-

sector datasets, 228 in all, of the 247 analysed. (The two antimatter datasets not shown are the $\bar{p}$ -He and $dd\mu^+$ entries, whose sectors have no counterpart row; this is why the table’s antimatter column totals 20 rather than the 22 quoted in Section 4.2.) The remaining 19 do not belong to a matter–antimatter sector pair at all: 16 are astrophysical bounds, quoted for a single sector with no laboratory antimatter counterpart, and the other three are the μ -N, $\bar{p}$ -He and $dd\mu^+$ entries, whose sectors have no counterpart row in the table.

4.4.2 Coverage by Dobrescu–Mocioiu Potential

Table 4.5 breaks the same comparison down by DM potential.

Potential	Matter datasets	Antimatter datasets	Status
V_{4+5}	58	3	Thin antimatter coverage
V_{9+10}	31	0	Gap
V_3	29	10	Best-covered potential (used throughout Section 4.3)
V_{1a}	14	0	Gap
V_2	17	3	Thin antimatter coverage
V_1	42	5	Thin antimatter coverage
V_{12+13}	16	0	Gap
V_{11}	7	0	Gap
V_8	5	1	Thin antimatter coverage
V_{15}	2	0	Gap
V_{16}	2	0	Gap
V_{2+3}	2	0	Gap (combination label)

Table 4.5: Matter versus antimatter dataset counts per Dobrescu–Mocioiu potential.

Seven of the twelve classified potentials, V_{11} , V_{15} , V_{16} , V_{1a} , V_{9+10} , V_{12+13} , and V_{2+3} , have *zero* antimatter-sector datasets. Only V_3 has antimatter coverage exceeding a handful of datasets, which is why fourteen of the fifteen matched pairs in Section 4.3 involve V_2 or V_3 .

Figures 4.17–4.21 show the dataset inventory and coverage matrix for the same database. They count *independent bounds* rather than dataset files: 48 of the 247 analysed datasets are the same measured bound filed under more than one coupling family, and are collapsed to a single curve before plotting, leaving 199 independent bounds. Row and column totals read off those figures are therefore systematically smaller

than the counts in Tables 4.4 and 4.5, which count files as the registry records them. The two views agree on every gap; they differ only in how multiply-filed bounds are counted.

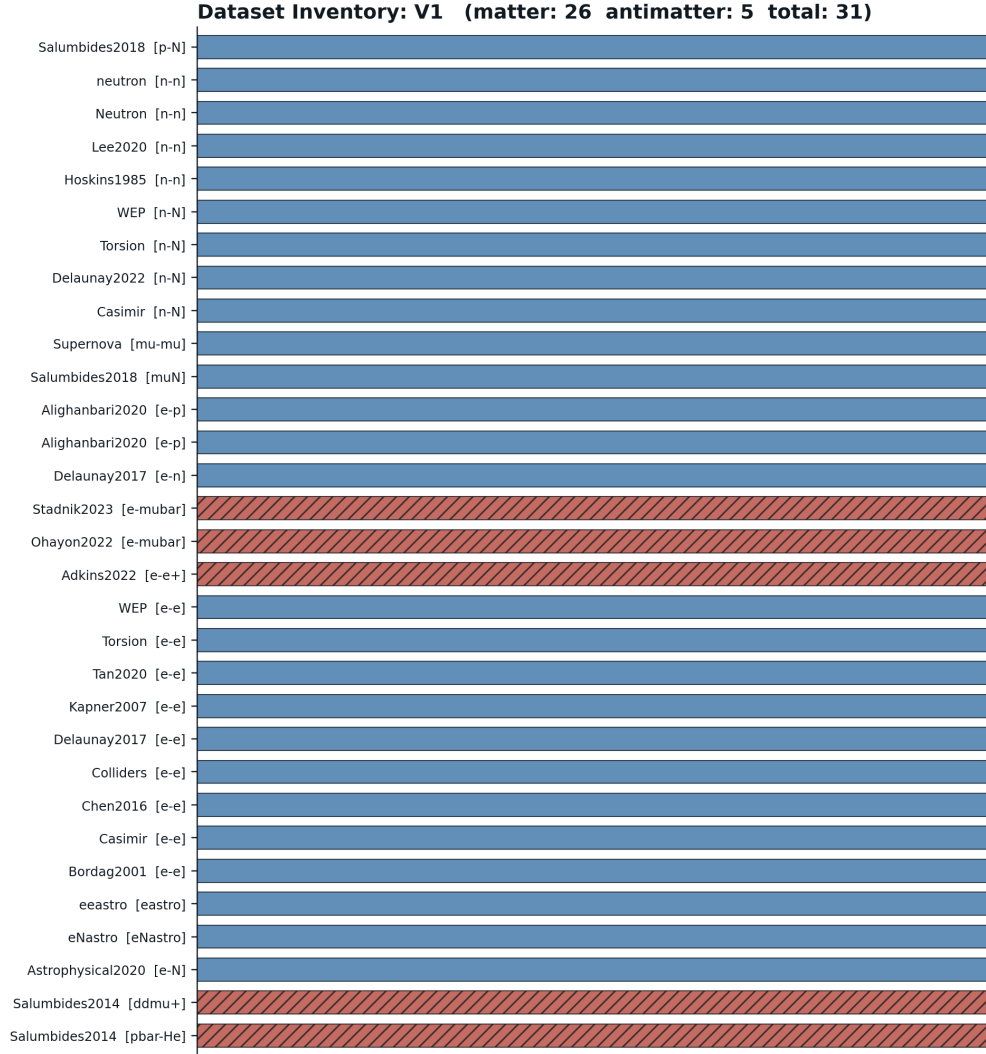


Figure 4.17: Dataset inventory for V_1 .

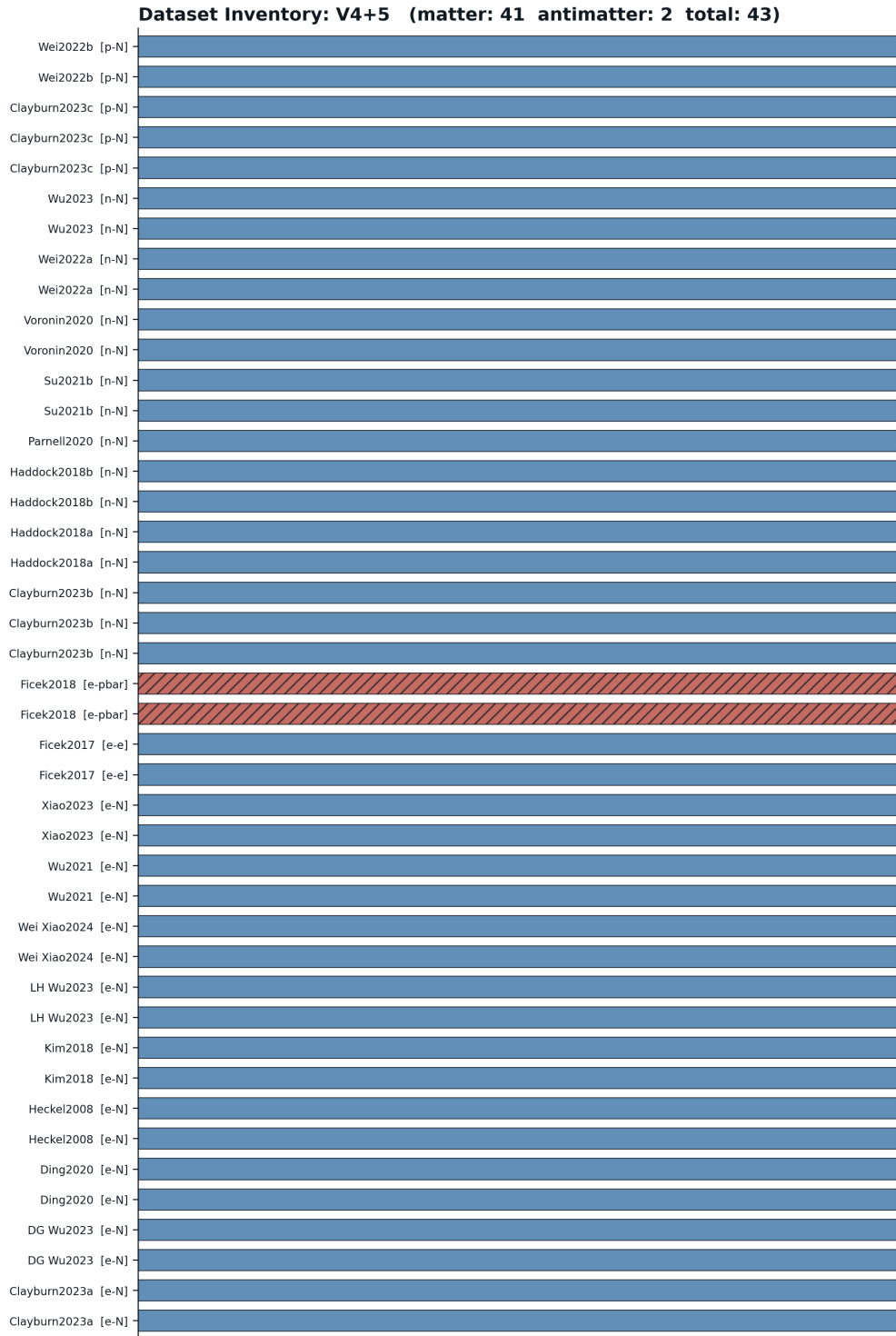


Figure 4.18: Dataset inventory for V_{4+5} , the most populated potential in the database (61 datasets).

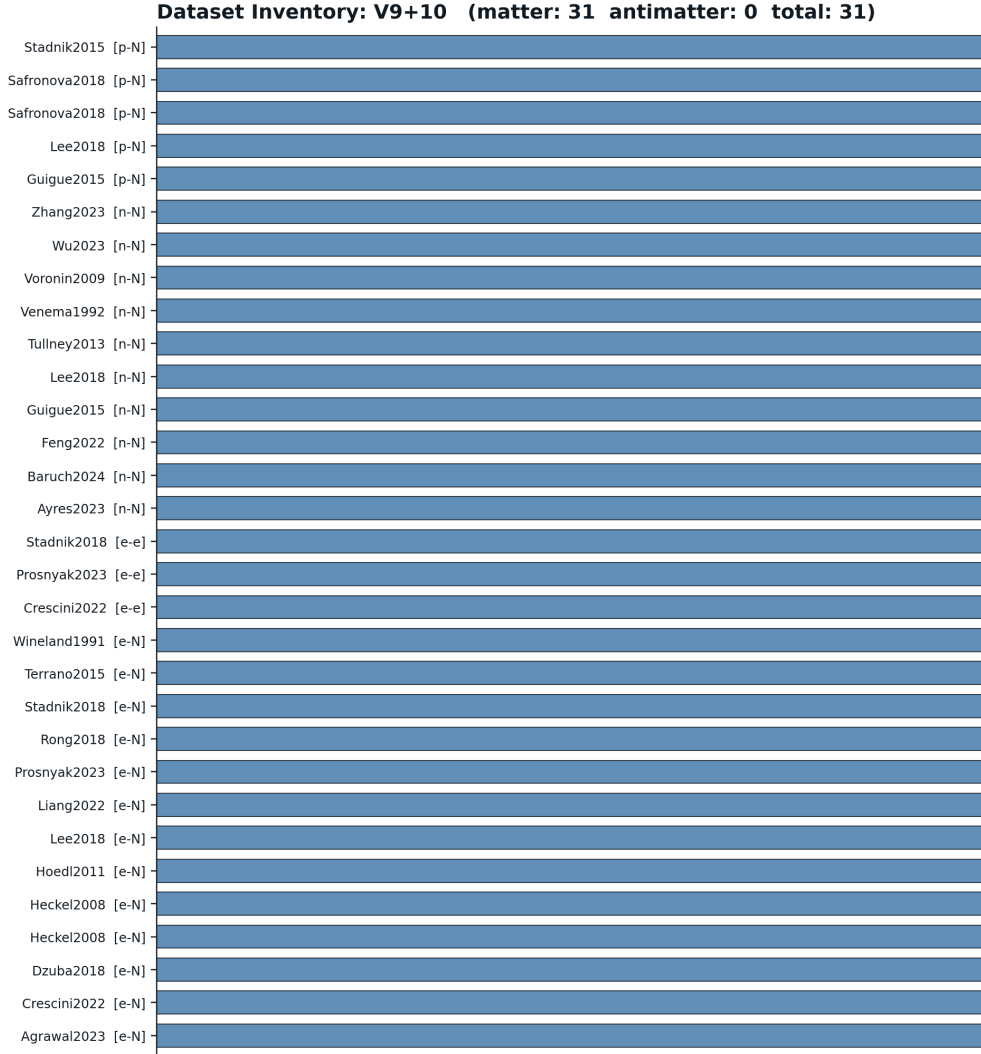


Figure 4.19: Dataset inventory for V_{9+10} .

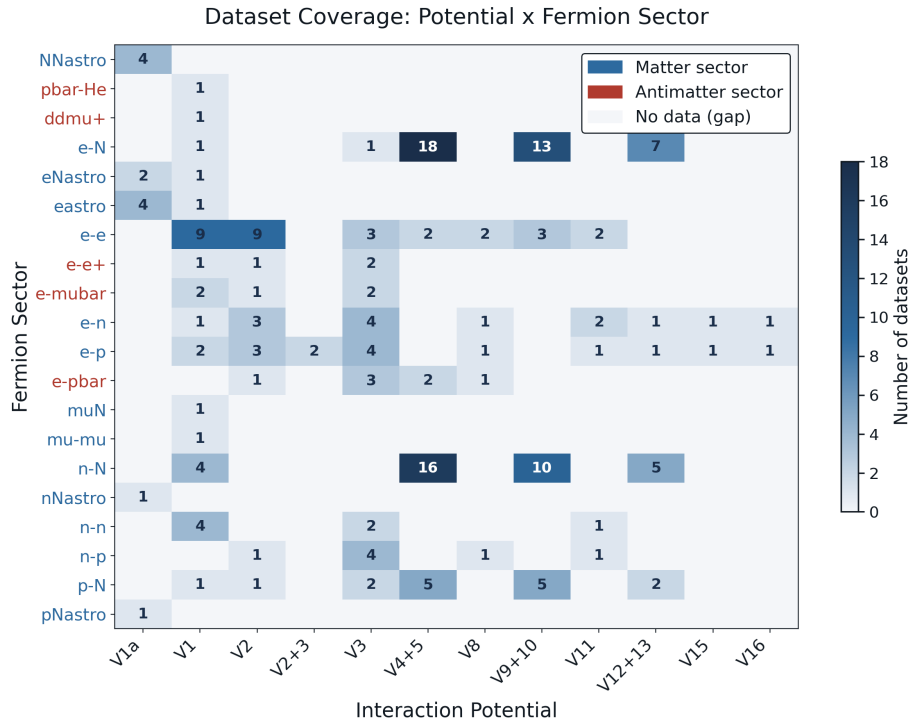


Figure 4.20: Dataset count by potential and sector; antimatter-sector rows are highlighted. Cells count independent bounds, not dataset files, so totals fall below those in Tables 4.4 and 4.5; the gaps identified are the same.

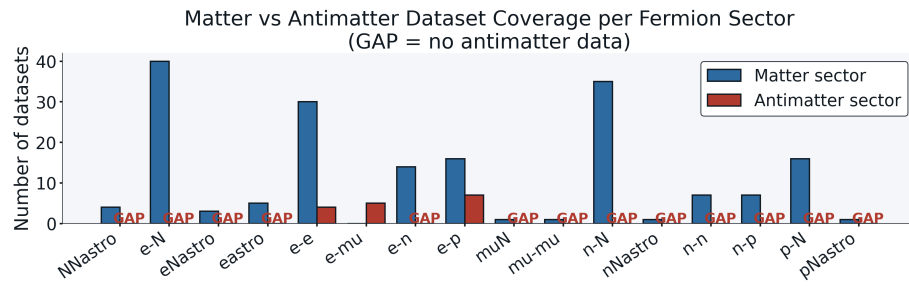


Figure 4.21: Matter-to-antimatter dataset count ratio by sector and potential.

4.4.3 Data-Quality and Framework Limitations

Two limitations of the current database and framework are distinct from the physics coverage gaps above and are stated here rather than worked around silently. First, the 36 datasets held out of the analysis (Section 4.2) are excluded rather than reclassified, and the reasons differ in kind. Twenty-eight are **Combined** envelope curves, twelve in $g_A g_V$ and sixteen in $g_p g_s$, which the source database marks **review_only** and assigns no potential; these are not a parsing gap that any filename convention could close, because no single V_n describes an envelope over several experiments, and they would have to be decomposed into their constituent bounds before they could enter a per-potential analysis. The sixteen $g_p g_s$ curves were identified only when every potential assignment in the database was validated against the source database’s curation records: a local filename prefix had caused them to be read as V_{1a} , which those records contradict. Two are contributor-template placeholders, and were previously being read as V_1 and V_2 datasets on the strength of a trailing digit in their filenames, which inflated those two potentials by one dataset each. The remaining one is a $g_p g_p$ astrophysical e - N bound: a real measurement, but one whose potential is unassigned upstream, and assigning it V_{1a} from the filenames of its siblings in other coupling families would be an inference rather than evidence. It is held out until the source publication settles it, and is the one exclusion that a single documentary check could reverse. The last five are the 90% CL Cong *et al.* (2025) curves, excluded as duplicates of the 95% CL set rather than as defective data; retaining both would count one experiment twice. Second, six entries under V_3 , V_{9+10} and V_{15} report interaction ranges that are physically implausible, reaching $\lambda \sim 10^{52}$ m against an observable universe of order 10^{26} m, almost certainly the result of a unit token being mis-detected during compilation. Three of the fifteen matched pairs in Table 4.3 draw on an affected dataset: the $g_A g_A \cdot V_3 \cdot e$ - e^+ pair on the matter side (Ficek *et al.*, 2017), and the two $g_A g_A \cdot V_3 \cdot e$ - $\bar{p}$ pairs that use the Ficek *et al.* (2018) bound on the antimatter side. None of these results is affected, because the asymmetry is evaluated only on the overlapping range of the two curves, and in every case the implausible tail lies outside that overlap: the ranges used are $[2.5 \times 10^{-12}, 1.6 \times 10^{-5}]$ m, $[2.0 \times 10^{-11}, 1.0 \times 10^{-8}]$ m and $[2.0 \times 10^{-11}, 1.3 \times 10^{-8}]$ m. A broader coverage claim drawn from the full database without this cleanup would nonetheless not be reliable.

4.4.4 Proposed Experimental Strategies

Three concrete directions follow from the coverage gaps above. First, future antimatter-sector measurement effort is best prioritised in the e - N , n - N , p - N , e - n , n - p , and n - n sectors, which together account for 150 of the 225 matter-sector datasets in the analysed database and zero antimatter-sector ones; precision antiproton experiments such as BASE, and antihydrogen spectroscopy such as ALPHA, are the nearest existing platforms for

producing bounds in these channels. Second, the same logic applies at the potential level to V_{1a} , V_{9+10} , and V_{12+13} , each with 14–31 matter-sector datasets and none on the antimatter side; V_1 , V_2 , and V_8 have thin but non-zero antimatter coverage and are better placed as candidates for *improving* existing precision rather than opening a channel from zero. Third, following the precedent documented by Cong *et al.* (2025), in which Wu *et al.* (2023) reanalysed comagnetometer data taken for sidereal Lorentz- and CPT-violation searches as new bounds on the V_{12+13} , V_{4+5} and V_{9+10} terms, a systematic search of existing BASE, ALPHA, and comagnetometer publications for data that could be reinterpreted within this framework is likely to close some of the identified gap faster, and at lower cost, than commissioning new measurements from scratch. Across all three directions, a new antimatter-sector bound is scientifically useful for a CPT test specifically when it approaches the precision of its matter-sector counterpart, not merely when it exists: a loose first measurement in an otherwise empty sector would simply reproduce the same sensitivity-gap-dominated $|A_\alpha| \approx 1$ pattern documented throughout this chapter, without resolving the CPT question it is meant to address.

Chapter 5

Conclusions

5.1 Summary

This thesis set out to close a specific gap: the Standard Model Extension (SME) and the Dobrescu–Mocioiu (DM) classification of exotic spin-dependent potentials describe the same physics in two languages, one relativistic and field-theoretic, the other non-relativistic and directly tied to laboratory observables, with no complete, systematic dictionary between them and no framework for using such a dictionary to compare matter- and antimatter-sector constraints as a test of CPT symmetry. Chapters 2 and 3 built the theoretical bridge, deriving the non-relativistic Hamiltonians generated by b_μ , $H_{\mu\nu}$, $d_{\mu\nu}$, and $g_{\lambda\mu\nu}$ via the Foldy–Wouthuysen transformation and matching each onto the DM potential catalogue V_1 – V_{16} . Chapter 4 applied that bridge to real data, compiling 283 experimental constraint datasets (247 analysed) into a single standardised database and computing the CPT asymmetry parameter A_α for every matter–antimatter pair the database could support, before turning the same database around to identify where it is currently silent and proposing concrete next steps. The result is not a single number or a single discovery, but a working pipeline: a mapping table connecting relativistic coefficients to measurable potentials, a standardised constraint database built to be extended, and a quantitative account of exactly how far current data can and cannot go toward testing CPT symmetry in the spin-dependent sector.

5.2 Revisiting the Objectives

All five specific objectives set out in Chapter 1 were achieved in full. The SME–DM mapping was derived completely for b_μ , $H_{\mu\nu}$, $d_{\mu\nu}$ and $g_{\lambda\mu\nu}$, reproducing the expected operator structure and CPT behaviour at every order in $1/m$ kept in this thesis. All three $d_{\mu\nu}$ components, $d_{i0} \rightarrow V_2$, $d_{ij} \rightarrow V_7, V_8$ and $d_{00} \rightarrow V_8$, match Kostelecký and Lane (1999a) exactly, including their signs, once that reference’s momentum-index convention

is applied consistently; an apparent sign discrepancy in the two momentum-dependent components was traced to that convention rather than to the derivation. A standardised database of 283 datasets was compiled across five experimental platforms, 247 of which entered the analysis. The CPT asymmetry parameter was defined and computed for all fifteen matchable matter–antimatter pairs, with the important qualification, itself one of this thesis’s principal findings, that A_α computed from one-sided experimental bounds cannot by itself distinguish genuine CPT violation from an ordinary sensitivity gap. The antimatter-sector coverage gap was quantified by both fermion sector and DM potential, and concrete experimental and archival strategies were proposed for closing it.

5.3 Principal Findings

The master matching table (Table 4.1) is this thesis’s central technical deliverable. Its single most physically significant result is the b_i derivation: a CPT-odd coefficient produces an exact sign flip between CPT-conjugate matter and antimatter states (at the same physical spin the two shifts coincide, as Kostelecký and Lane’s antiparticle rule requires), obtained independently from the Lagrangian’s general CPT transformation and from an explicit hole-theory derivation of the antiparticle Hamiltonian, after a charge-conjugation argument on the bilinear alone was checked and found insufficient to produce the sign flip on its own (Section 4.1.1). The most conceptually important result, however, is methodological rather than numerical: the asymmetry parameter $A_\alpha = (g_f - g_{\bar{f}})/(g_f + g_{\bar{f}})$, as actually computed from published experimental limits, cannot distinguish a genuine CPT-violating signal from an ordinary difference in sensitivity between the matter- and antimatter-sector experiments being compared. An exact signed CPT-odd relation makes the formula diverge rather than saturate at 1; what actually drives real output to $|A_\alpha| \approx 1$ is the ordinary behaviour of a ratio of two independent, always-positive upper bounds of very different tightness, a pattern that arises regardless of the true CPT parity of the underlying physics. This was confirmed empirically: the spin-dependent $g_A g_A$ pairs and the spin-independent $g_s g_s$ pair alike showed statistically significant asymmetries after correcting for the correlated nature of interpolated constraint points, yet none can currently be read as evidence of CPT violation rather than sensitivity-gap saturation. Of the 247 datasets analysed, only 22 involve an antimatter sector, and from these only fifteen matter–antimatter pairs could be matched; seven of twelve classified DM potentials and six of eleven fermion-sector pairs have zero antimatter-sector coverage despite substantial matter-sector data. This is the binding constraint identified by this thesis: not a lack of matter-sector precision, but an almost complete absence of comparably precise antimatter-sector measurements in exactly the channels where matter-sector data is most abundant.

5.4 Limitations

Two limitations bear on how much weight the results above can be given. Thirty-six of the 283 compiled datasets (roughly one in eight) are held out of the analysis rather than classified, twelve of them source-flagged **review_only** envelope curves to which no single potential applies; and a small number of entries report physically implausible interaction ranges likely caused by a unit-detection error. Neither affects the fifteen matched pairs analysed, but the unit error should be resolved before a broader coverage claim is drawn from the full database. Finally, this thesis derives the non-relativistic limit of four of the eight SME coefficients classified in Chapter 2. Those four are not an arbitrary subset: they are the complete set that generates spin-dependent non-relativistic potentials, the remaining coefficients contributing only spin-independent terms and therefore having no image in the Dobrescu–Mocioiu basis. The derivations are nonetheless carried only to the orders in $1/m$ retained here, and the higher-order terms of the same coefficients are not represented in the asymmetry analysis.

5.5 Recommendations for Future Work

The most direct theoretical extension is to carry the existing four reductions to higher order in $1/m$, where the spin-dependent sector acquires further structure, and to derive the spin-*independent* coefficients a_μ , $c_{\mu\nu}$ and e_μ . The latter fall outside the Dobrescu–Mocioiu basis used here but bear on the scalar V_1 channel, and the kinetic-sector field redefinition already required for $d_{\mu\nu}$ and $g_{\lambda\mu\nu}$ carries over directly to $c_{\mu\nu}$, which occupies the same position in the modified kinetic operator. On the database side, decomposing the twenty-eight **Combined** envelope curves ($g_A g_V$ and $g_p g_s$) into their constituent per-potential bounds, and settling the one unassigned astrophysical bound against its source publication, are concrete and bounded cleanup tasks. The e - μ sector needs new data rather than cleanup: every current bound there comes from muonium and so sits on the antimatter side, with no matter-sector counterpart. Experimentally, new antimatter-sector bounds should be prioritised in the e - N , n - N , p - N , e - n , n - p , and n - n sectors and in the V_{1a} , V_{9+10} and V_{12+13} potentials, and existing BASE, ALPHA, and co-magnetometer publications should be searched for data that could be reinterpreted within this framework before new measurements are commissioned. Throughout, a new antimatter-sector bound is useful for a CPT test specifically when it approaches the precision of its matter-sector counterpart, not merely when it exists.

5.6 Closing Remarks

The gap this thesis set out to address, a missing dictionary between the SME and the Dobrescu–Mocioiu potentials, and no systematic way to use that dictionary to compare matter- and antimatter-sector constraints, is now substantially, though not completely, closed. The mapping table, the compiled 283-dataset database, and the gap analysis together give a reproducible, extensible pipeline rather than a fixed set of numbers: the database is built to absorb new datasets as they are published, and the pairing and asymmetry-analysis methodology apply unchanged to any new antimatter-sector measurement that lands in one of the gap channels identified here. The most durable contribution of this thesis may not be any single derived Hamiltonian or compiled bound, but the demonstration that an asymmetry parameter built from one-sided experimental bounds is a sensitivity-gap diagnostic before it is a CPT diagnostic, and that closing the antimatter-sector coverage gaps mapped in this thesis is a precondition for that parameter to mean what it was designed to mean.

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Appendix A

Notation and Symbol Reference

This appendix collects, in one place, every symbol, operator, and sign convention used across Chapters 1–5.

A.1 Dirac Algebra

Symbol	Meaning	Context
γ^μ	Dirac gamma matrices ($\mu = 0, 1, 2, 3$)	Relativistic fermion field theory
γ^0	Time-like gamma matrix	Block-diagonal: diag(1, 1, −1, −1) in the Dirac representation
γ_5	$i\gamma^0\gamma^1\gamma^2\gamma^3$	Chiral matrix; off-diagonal in the Dirac representation
β	γ^0	Used throughout the Foldy–Wouthuysen expansion
α^i	$\gamma^0\gamma^i$ ($i = 1, 2, 3$)	Velocity operator; off-diagonal in the Dirac representation
Σ^i	diag(σ^i, σ^i)	4×4 spin matrix
σ^i	Pauli matrices ($i = 1, 2, 3$)	2×2 spin operators, non-relativistic limit
$\boldsymbol{\sigma}$	$(\sigma^1, \sigma^2, \sigma^3)$	Pauli vector
D_μ	Gauge-covariant derivative	Appears in the general SME Lagrangian (Ch. 2)
η_Γ	CPT parity of a bilinear $\bar{\psi}\Gamma\psi$	+1 for $\Gamma \in \{\mathbb{1}, \gamma^\mu, \sigma^{\mu\nu}\}$, −1 for $\Gamma \in \{\gamma_5, \gamma_5\gamma^\mu\}$ (Ch. 2)

Table A.1: Dirac algebra notation used in Chapters 2–4.

Symbol	Type	CPT	Lorentz	Definition / leading role
a_μ	Vector	Odd	Odd	$\mathcal{L}_a = -a_\mu \bar{\psi} \gamma^\mu \psi$; no spin-dependent leading term
b_μ	Vector	Odd	Odd	$\mathcal{L}_b = -b_\mu \bar{\psi} \gamma_5 \gamma^\mu \psi$
b_i	Spatial part of b_μ	Odd	Odd	Generates $-\mathbf{b} \cdot \boldsymbol{\sigma}$ (matter)
b_0	Temporal part of b_μ	Odd	Odd	Generates $-b_0(\boldsymbol{\sigma} \cdot \mathbf{p})/m$
$c_{\mu\nu}$	Tensor, kinetic sector	Even	Odd	Direction-dependent kinetic modification
$H_{\mu\nu}$	Antisymmetric tensor	Even	Odd	$\mathcal{L}_H = -\frac{1}{2} H_{\mu\nu} \bar{\psi} \sigma^{\mu\nu} \psi$
H_{ij}	Spatial part of $H_{\mu\nu}$	Even	Odd	Magnetic-like; generates $+\mathcal{H}_B \cdot \boldsymbol{\sigma}$
H_{0i}	Mixed part of $H_{\mu\nu}$	Even	Odd	Electric-like; generates $+\frac{1}{m} \boldsymbol{\sigma} \cdot (\mathbf{p} \times \mathbf{H}_E)$
$d_{\mu\nu}$	Traceless tensor, kinetic sector	Even	Odd	$\mathcal{L}_d = \frac{1}{2} i \bar{\psi} d^{\mu\nu} \gamma_5 \gamma_\mu \overleftrightarrow{\partial}_\nu \psi$
d_{i0}	Mixed part of $d_{\mu\nu}$	Even	Odd	Generates $+d_{i0} m \sigma^i$ (m^1 enhancement)
d_{ij}	Spatial part of $d_{\mu\nu}$	Even	Odd	Generates $+d_{ij} p^j \sigma^i$ (velocity-dependent)
d_{00}	Temporal part of $d_{\mu\nu}$	Even	Odd	Generates $+d_{00}(\boldsymbol{\sigma} \cdot \mathbf{p})$
e_μ, f_μ	Dimensionless vectors	Odd	Odd	No spin-dependent term: e_μ is spin-independent, f_μ absent at linear order
$g_{\lambda\mu\nu}$	Dimensionless tensor	Odd	Odd	Kinetic-sector spin couplings; generates V_2, V_7, V_8

Table A.2: SME fermion-sector coefficients used in this thesis.

A.2 SME Coefficients

A.3 Dobrescu–Mocioiu Coupling Constants

Symbol	Meaning
g_s	Scalar coupling constant
g_p	Pseudoscalar coupling constant
g_V	Vector coupling constant
g_A	Axial-vector coupling constant

Table A.3: Dobrescu–Mocioiu coupling constants.

A.4 The Asymmetry Parameter

The CPT asymmetry parameter is defined, for interaction channel $\alpha \in \{s, p, V, A\}$, as

$$A_\alpha = \frac{g_\alpha^f - g_\alpha^{\bar{f}}}{g_\alpha^f + g_\alpha^{\bar{f}}}, \quad (\text{A.1})$$

where g_α^f is the effective coupling constant for fermion f in channel α and $g_\alpha^{\bar{f}}$ is the corresponding coupling for its antifermion. Under exact CPT symmetry, $A_\alpha = 0$ for every channel. For a genuinely CPT-odd, exactly signed coupling ($g_\alpha^{\bar{f}} = -g_\alpha^f$), the denominator vanishes identically, so A_α diverges rather than saturating at 1 (Chapter 4). SPINDEP’s actual inputs, g_α^f and $g_\alpha^{\bar{f}}$, are independent, always-positive experimental upper bounds, not signed measured couplings; a ratio of two independent positive bounds of very different tightness approaches ± 1 whenever one bound is far tighter than the other, regardless of the true CPT parity of the underlying physics.

A.5 Statistical and Database Notation

Symbol	Meaning
<code>{V}{Author}{Year}_m_abs_{sector}.csv</code>	Filename convention for a compiled constraint dataset
ee, eebar, ep, epbar, emu, emubar, ...	Fermion-sector labels; bar -suffixed sectors are antimatter
UNKNOWN	Potential label the parser could not resolve from a filename; such datasets are compiled into the registry but held out of the analysis
$g_A g_A, g_p g_s, g_s g_s, g_V g_V, g_A g_V, g_p g_p$	Coupling-family classification tags for a dataset
χ^2	Chi-squared statistic
dof_{eff}	Effective degrees of freedom, corrected for autocorrelation among interpolated grid points
Z_{eff}	Weighted- χ^2 significance after rescaling χ^2 by $\text{dof}_{\text{eff}}/n$ and testing on dof_{eff} degrees of freedom, as a one-sided Gaussian equivalent
$\overline{ A_\alpha }$	Mean absolute asymmetry across the grid for one matched pair
95% CI	Bootstrap (2000 resamples) confidence interval on $\overline{ A_\alpha }$

Table A.4: Statistical and database notation used in Chapter 4.

Appendix B

Computational Resources

The symbolic and numerical work underlying Chapters 3 and 4 is implemented as executable code rather than reproduced in full here, following the convention that large or actively maintained programs are referenced by location rather than transcribed into a printed appendix. The principal resources are:

- **SPINDEP** – the dataset parser, unit-conversion routines, matter–antimatter pairing logic, χ^2 /bootstrap statistical pipeline, constraint-atlas and gap-analysis plotting code, and the compiled 283-dataset registry described in Chapter 3 (247 of which enter the analysis), maintained as a version-controlled Python package. The full analysis in Chapter 4 is reproducible from a single command, `spin run -data ./datasets`, part of a ten-command command-line interface (`spin run`, `test`, `validate`, `import`, `gaps`, `atlas`, `config`, `batch`, `info`, `start`) that wraps the pipeline for use without editing any code; `spin start` additionally launches a browser-based interface for interactive exploration of the compiled database and per-pair results. The core parsing, matching, and statistical modules are covered by an automated test suite (130+ tests, run with `pytest`), added to guard the numerical results in Chapter 4 against silent regression as the codebase and the compiled dataset continue to be extended after this thesis.
- **Foldy–Wouthuysen derivation notebooks** – executable SymPy notebooks (`FW_bmu_term.ipynb`, `FW_Hmunu_term.ipynb`, `FW_dmunu_term.ipynb`, `FW_gmunu_term.ipynb`) reproducing, symbolically, every non-relativistic Hamiltonian derived in Chapter 4, together with supporting modules for Dirac algebra and Pauli-matrix manipulation, and the symbolic limit calculation used to establish the divergence of the asymmetry parameter (Eq. (4.7)) under an exact signed CPT-odd substitution. A further notebook, `FW_antiparticle_all_coefficients.ipynb`, derives the matter–antimatter relation for every SME coefficient family, both between CPT-conjugate states and at fixed physical spin, and checks it against each coefficient’s CPT parity and against Kostelecký and Lane’s (1999b) antiparticle rule.

- **Theory notes** – worked derivations in prose form for the b_μ , $H_{\mu\nu}$, $d_{\mu\nu}$, and $g_{\lambda\mu\nu}$ terms and the resulting potential-matching table, cross-referenced against the notebooks above.
- **Constraint datasets** – the underlying two-column CSV files for all 283 compiled datasets, of which 247 enter the analysis, organised by coupling family and interaction class as described in Chapter 3, Section 3.2.1.

The constraint-atlas figures (Figures 4.1–4.13), the matter–antimatter comparison plot (Figure 4.14), and the coverage figures (Figures 4.17–4.21) in Chapter 4 are generated directly from this codebase and are reproducible by re-running the corresponding SPINDEP plotting routines against the compiled dataset registry.

Appendix C

Supplementary Coverage Figures

Section 4.4 presents the per-potential dataset-inventory figure for three representative potentials, V_1 , V_{4+5} , and V_{9+10} , chosen to illustrate the thin-coverage, dense-coverage, and zero-antimatter-coverage cases respectively. The same figure exists for every other classified potential and is included here in full for completeness, without repeating the discussion already given in Section 4.4; the counts underlying each panel are also already tabulated in Table 4.5.

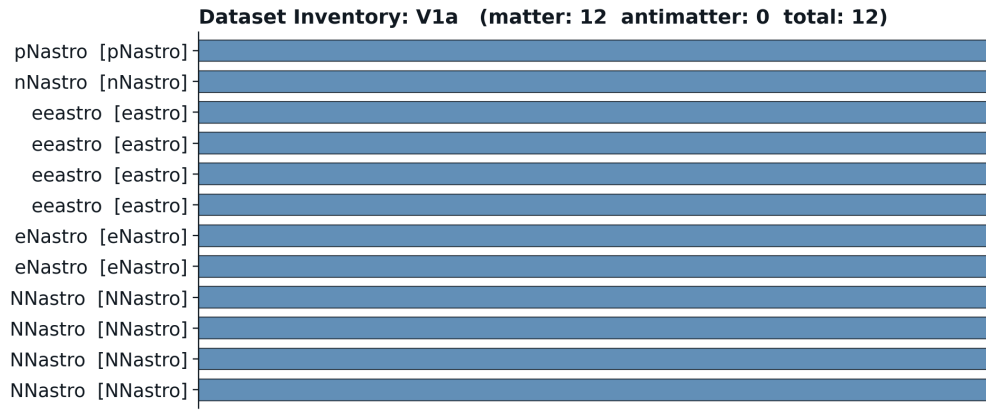


Figure C.1: Dataset inventory for V_{1a} .

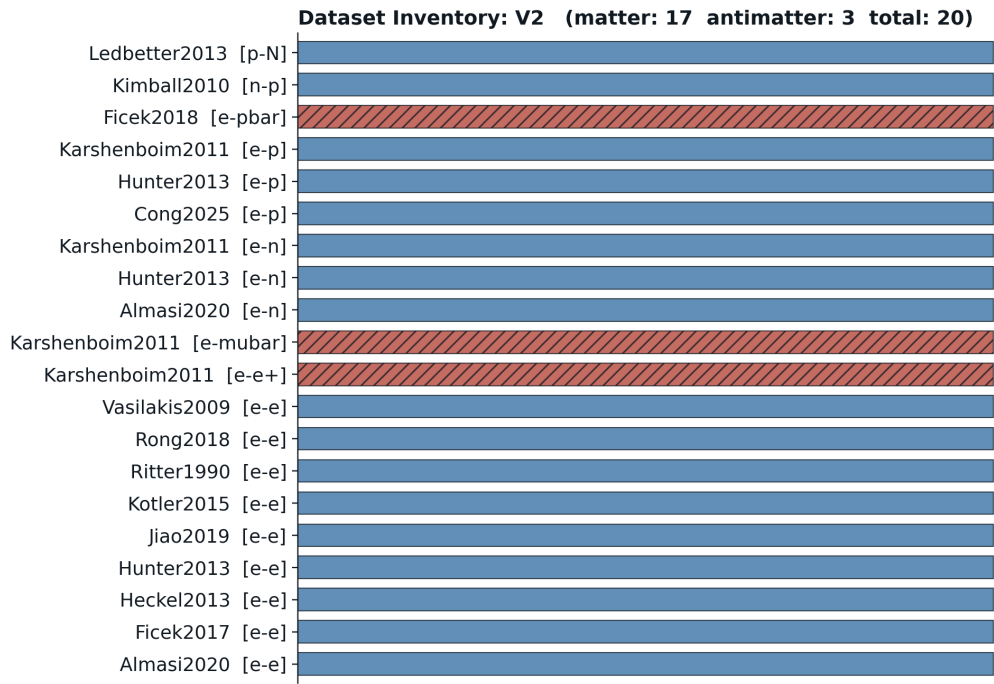


Figure C.2: Dataset inventory for V_2 .

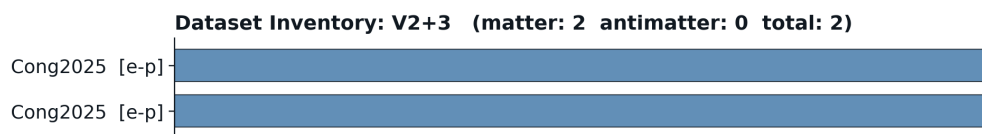


Figure C.3: Dataset inventory for V_{2+3} .

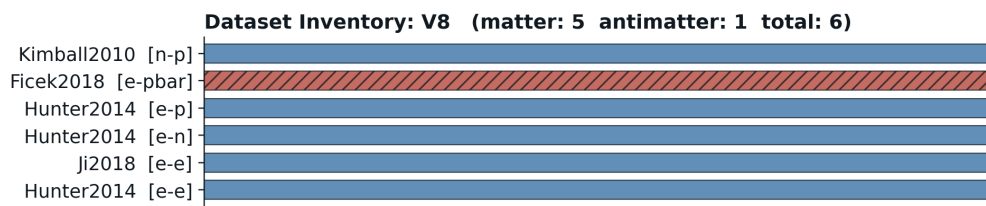


Figure C.4: Dataset inventory for V_8 .

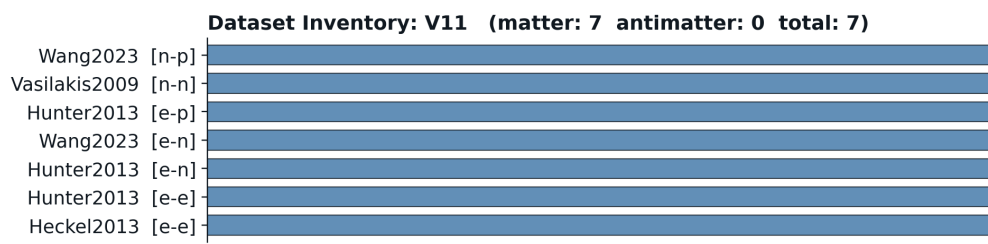


Figure C.5: Dataset inventory for V_{11} .

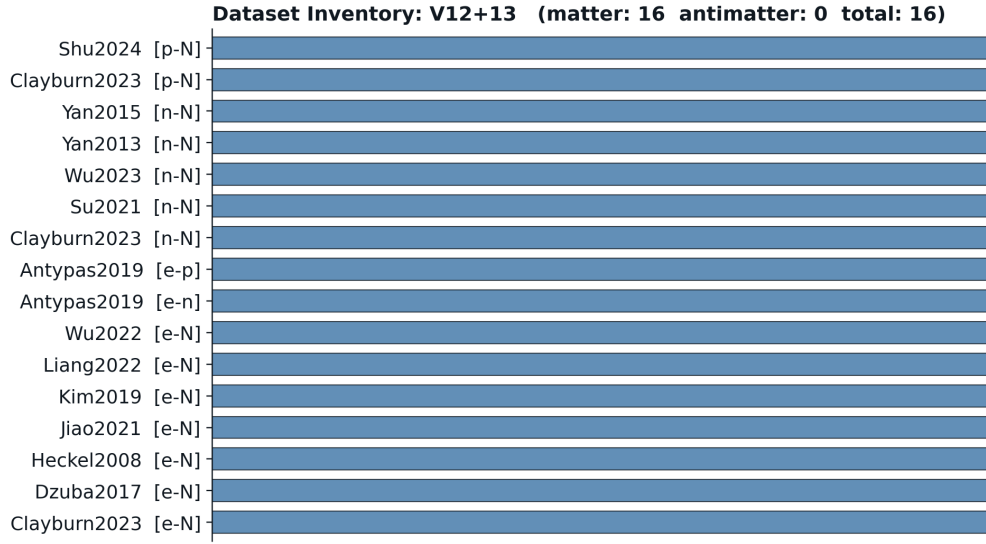


Figure C.6: Dataset inventory for V_{12+13} .

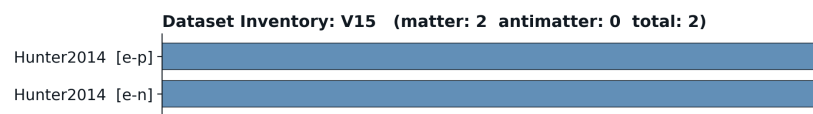


Figure C.7: Dataset inventory for V_{15} .

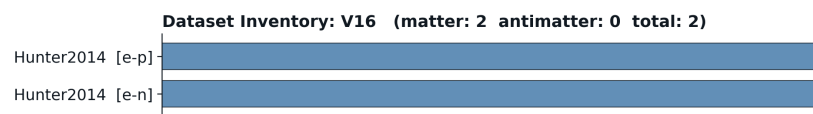


Figure C.8: Dataset inventory for V_{16} .